

# Social media meets FinTech platforms: How do online emotions support credit risk decision-making?

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## ABSTRACT

As emerging FinTech platforms face pressure in efficiently managing credit risk, the human emotional spectrum of FinTech platform borrowers within social media becomes a potential source for gaining insight into and evaluating their financial behaviors. Collaborating with an Asian FinTech platform, we investigate the impact of social media emotions on a platform's loan-approval decisions and repayment-reminder interventions before due dates. We demonstrate that anger at the pre-approval stage has a U-shaped relationship with platform borrowers' default probability. We reveal what we call "*a bright side of anger*" with respect to curbing financial credit risk: moderate intensity of anger at the pre-approval stage suggests a lower loan default probability. We also find that the average happiness tendency of platform delinquent borrowers' at the pre-maturity stage becomes informative and valuable, as it shows a U-shaped relationship with loan default; as for anger, it does not work therein. Furthermore, our field experiment indicates that a positive-expectation reminder is useful for prompting repayment when delinquent borrowers are in strong emotional intensities, regardless of anger or happiness. However, a negative-consequence reminder results in a higher default probability for delinquent borrowers who maintain high immediate happiness before the loan maturity dates. We draw on the classical appraisal theory of emotions and the feelings-as-information theory to interpret our findings. We offer non-trivial theoretical and practical implications to support FinTech platform credit risk decision-making by investigating the value of social media emotions and advocating for cross-functional coordination between debt approval and debt collection departments.

## 1. Introduction

Credit risk management has always been one of the utmost critical managerial challenges for the micro-finance industry; indeed, FinTech platforms endeavor by many and various means to control and minimize credit risks in their lending services. The literature, correspondingly, offers an abundance of approaches to explain the factors underlying credit risks. Most of them, in general, have focused on objective features, such as loan attributes (e.g., loan amount, loan term, interest rate), borrowers' characteristics (e.g., demographics, socioeconomic status) [1] and social capital [2,3]. Some recent studies

have paid attention to textual information and usage of loan narratives to infer borrowers' financial hardship [4]. That said, only a limited number of studies have explored credit risk behavior through the lens of borrowers' subjective features. Credit risk, typically manifested as default behavior, is a type of social deviation [5]. It is more likely to be committed intentionally and to be driven by subjective thoughts (about inner feelings/strains and external circumstances) than occasional neglect of the deadline [6]. Thus, applying subjective features can not only supplement credit risk assessment but, more importantly, can also help to explain borrowers' psychological mechanisms in credit risk behavior [5].

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External information (e.g., extracted from online social media) can fairly state users' subjective features [7–10]. Effective natural language processing (NLP) techniques enable us to extract various kinds of information from users' social media [11]. While the literature has widely documented the value of social media information in explaining users' subjective traits and behaviors, such as personality [12] and the onset of depression [13], it has hardly focused on understanding borrowers' credit risks for more informed loan approval decisions. A small proportion of studies have documented how social media information can be leveraged to manage credit risks, but they have explored social media usage/disclosure behavior [14] or social network structure [15], and as such, evidence of the value of extracted subjective features in credit risk assessment and decision-making for loan approvals has been scant.

Moreover, social media possesses a unique advantage in supporting decision-making in credit risk management: it enables us to track an individual borrower's subjective status over time, in both pre- and post-loan-application/approval periods. More importantly, online social media, with minimal privacy concerns, allows FinTech platform practitioners 1) to understand borrowers' follow-up subjective status after loan granting and around the repayment due time, and thus also, 2) to implement effective proactive repayment interventions. The extant literature on repayment interventions generally has employed a uniform strategy for all borrowers in a platform [16], rarely taking borrower heterogeneity into account. By utilizing borrowers' social media information, we aim to achieve two goals relevant to decision-making in credit risk management: effective credit risk assessment for a platform's loan-approval decisions, and powerful credit interventions by the platform for prompting timely repayment.

In particular, we focus on social media emotions—the affective states borrowers express in their online posts—and draw from two complementary theoretical lenses to justify their potential. First, *appraisal theory of emotions* [17,18] elucidates why borrowers experience certain emotions, such as anger or happiness, by proposing that individuals continually evaluate events or information (e.g., daily life events, platform reminders) with respect to their goals, well-being, or sense of control. Second, the *feelings-as-information theory* [19,20] clarifies how these emotional states, once formed, serve as internal cues guiding subsequent decisions, particularly under uncertainty. Prior work in other sectors has consistently shown that emotions, such as anger or happiness, can predict behaviors ranging from risk-taking [21] to consumption choices [22]. In the context of fulfilling financial obligations—a behavior closely tied to moral and prosocial dimensions [16]—emotional states may signify how borrowers assess the feasibility of repayment and ultimately make their final repayment decisions.

In a nutshell, this study attempts to fully investigate the role of social media emotions in FinTech platform credit risk management by empirically answering the following questions: (1) *which emotion types of FinTech platform borrowers are more useful for the assessment of credit risks?* And, (2) *how do borrowers with different emotion types and intensities respond to reminder interventions aiming to prompt repayment?* By addressing these questions, we can demonstrate the practical value of incorporating emotional data into decision-making processes within FinTech platforms, thereby improving risk assessment, loan approval decisions, and repayment-reminder intervention strategies.

To answer these questions, we leveraged both secondary data and a field experiment design, employing text-mining techniques. Specifically, we collaborated with a leading Asian FinTech platform to obtain granular information on borrowers' repayment status, demographics, and loan attributes. We focus on only borrowers who were continuously using online social media throughout the loan process covering both the pre- and post-loan-application periods. The focal platform provided us with concrete information on borrowers' social media presence (e.g., number of fans, number of “likes” received) and posts. We apply machine learning-based text-mining techniques to extract diverse

emotion types from the social media posts, and empirically explore whether and what borrowers' emotion types are precursors of their loan defaults. Then, we conduct a field experiment for tuning of treatments by positive-expectation and negative-consequence repayment-reminder intervention strategies in order to examine the repayment responses of borrowers with distinct emotions at the loan pre-maturity stage. Note that, as discussed above, because borrowers' social media emotions are influenced by external circumstances and might be altered frequently, to make empirical results more robust and/or offer more insights, we measure any type of emotion (e.g., happiness, anger) as an immediate psychological status that occurs at a short term (we coin it *immediate emotion*) or as an average psychological status that occurs over a relatively long term (i.e., *average emotional tendency*) [23]. Intuitively, borrowers' immediate emotions and average emotional tendencies may both correlate with credit risk behavior, either in the same or different directions. Immediate emotions are experienced temporally closer to the time point of borrowers' decision-making (e.g., pre-approval dates and pre-maturity dates), while average emotional tendencies reflect borrowers' long-term overall (and more likely genuine) emotional status.

In this paper, we have focused primarily on two common and representative social media emotions, happiness and anger, given the high pairwise correlations between positive and negative emotions, and the fact that both happiness and anger demonstrate higher values and higher variances relative to other emotions, manifesting themselves as salient candidates for having an association with borrowers' payment performance. Our empirical analyses support all of our hypotheses, which show distinct findings regarding the value of social media emotions at the pre-approval and pre-maturity stages in evaluating credit risks, supporting decision-making of FinTech platform managers. Specifically, we find that anger at the pre-approval stage, regardless of whether it is measured as an immediate reaction or average tendency, presents a significant U-shaped relationship with borrowers' default probability; happiness is ineffective at the pre-approval stage. Note that whereas intuition drives most of us to think of the negative relationship between anger and moral or prosocial performance [24], we reveal what we call “a bright side of anger” in terms of the curbing of financial credit risk: a moderate intensity of anger at the pre-approval stage suggests lower loan default probability. However, when we consider the relationships between eventual default behavior and emotions at the loan pre-maturity stage for delinquent borrowers (i.e., those who fail to fully repay their loans on time), we find that average happiness tendency becomes more informative: it also has a U-shaped relationship with loan default; anger does not work therein. In addition, our field experiment discloses some emotion-related conditions that would impact the effectiveness of well-designed repayment-reminder intervention strategies. In general, a positive-expectation reminder is useful to prompt repayment when delinquent borrowers are in a state of strong emotional intensity, regardless of anger or happiness. However, a reminder message conveying negative consequence deterrence would result in a higher default probability of borrowers who maintain higher immediate happiness before the loan maturity dates.

We expand the extant literature in four respects. *First*, to the best of our knowledge, we are among the first to investigate the value of social media emotions in the evaluation of FinTech platform borrowers' credit risks. Diverse emotion types draw a holistic picture of borrowers' psychological status [25], and we uncover some useful emotion types for credit risk assessment. Particularly, we analyze how social media emotions relate to default by differentiating two emotion measurements: immediate emotion and average emotional tendency. We demonstrate their complementary value in credit risk assessment to make appropriate FinTech platform decisions. *Second*, while the literature on FinTech platform credit risk evaluation mostly attends to features pertinent to the pre-approval stage [26], we leverage online social media to additionally explore how borrowers' most recent emotions at the loan pre-maturity stage would accompany default behavior. This

analysis has multifold merits: it enables us to utilize and compare the roles of psychological features at different stages in reinforcing credit risk assessment; it advocates for cross-functional coordination between debt approval and debt collection departments; it offers a potential avenue by which to design more delicate operational interventions, and extends the value of social media to the micro-finance industry. All of these merits facilitate a comprehensive understanding of borrowers' dynamic psychological status and credit behaviors during a loan process, with the least data privacy concerns. *Third*, we contribute to the FinTech platform credit risk literature by examining how borrower emotions moderate the effectiveness of different repayment-reminder strategies (e.g., positive-expectation vs. negative-consequence). Our empirical evidence confirms that the impact of such reminders varies significantly depending on a borrower's emotional intensity and emotional type. This step beyond baseline risk assessment, i.e., tailoring repayment interventions to distinct emotional profiles, demonstrates the potential for more personalized, targeted credit risk management. Practically, our study provides FinTech practitioners with novel insights into leveraging social media-based emotional data for more accurate credit risk management, including both initial approval decisions and subsequent repayment reminders. By constructing dynamic, subjective, emotion-related features, platforms can enhance risk assessment decision-making efficiency and design more impactful, personalized interventions.

## 2. Literature review

This paper stems from the following two streams in the literature: credit risk management in FinTech platforms and the relationships between emotions and subsequent decision-making.

### 2.1. Strategies to improve credit risk decision-making in FinTech platforms

First, the literature has endeavored to identify useful features for improving the accuracy of credit risk assessment and loan-approval decisions. Commonly-adopted features include loan attributes and borrower characteristics. With regard to loan attributes, for example, the larger a loan size or interest rate is, the higher the default probability is [27–29]. [1] find that loan purpose also plays an important role in predicting borrowers' default risks. Regarding borrower characteristics, [30] document that hard information, which refers to structured and quantifiable borrower information such as age, gender, and income level, contributes to the evaluation of default risks [31,32]. Scholars also document that borrowers' social connections help predict loan defaults [2,3,33,34]. Recently, more and more studies have used and disclosed the value of features from nontraditional sources in credit risk prediction. Popular alternative sources include mobile communication carriers (for cellphone usage and mobility traces features), e-commerce platforms (for shopping activities features), and online social media (for social media presence features) [35–37]. For example, [38] reveal that both cellphone usage and shopping activities features present superior predictive power to traditional demographic features in microloan credit risk assessment. [14] show that borrowers' activeness in social media and their self-disclosure of social media accounts can improve the prediction of their default probabilities.

Other studies seek features from unstructured soft information to bolster credit risk evaluation accuracy. For example, textual contents strategically narrated by borrowers from the product (loan or project) description helps to predict microloan default [4] and funding success in crowdfunding [39]. By contrast, external information sources such as online social media tend to reveal borrowers' subjective features in a more unbiased way [9,40]. [40] find that the sentiments of enterprises' social media posts help to predict their future credit risks. [9] demonstrate that the certain opinion-based constructs derived from social media posts can be indicative of their financial behaviors and

creditworthiness. Besides, some scholars, extracting borrowers' personality traits from their social media posts, reveal that extroversion, conscientiousness, neuroticism, and openness are significantly related to crowdfunding projects' funding level [41]. [4] suggest that extroversion is highly associated with loan credit risks. [42] document that people with a negative personality are more prone to default.

Other than credit risk assessment, prompt post-loan management is also crucial for mitigating losses from defaults. A post-loan management arsenal includes repayment-reminder strategies and loan recovery strategies. [16] explore two kinds of repayment reminders, and suggest that although a message delivering negative consequence deterrence has an equivalent effect to a positive prosocial compliance message on default prevention, its effect does not endure. Regarding loan recovery strategies, scholars examine and validate the effectiveness of extending financial incentives (e.g., a waiver on interest and delay fees) [43] and social shaming [44,45] for prompting loan repayment. Other studies also developed optimization algorithms to improve debt collection efficiency [46–48].

Despite that prior studies have developed miscellaneous features to facilitate credit risk assessment, those focusing on subjective traits are limited in both the number and scope. [5] suggest that loan defaulting should be considered to be financially deviant behavior because default behavior violates not only the social norm that one shall repay a debt but also the legal contract between a borrower and a lender [49]. Deviant behavior is driven by psychological reasons such as strains and negative effects [6]. [5] construct a theoretical framework to connect several psychological features (e.g., economic pressure, difficulty in socialization, inferior feeling) with loan default. Similarly, [9] introduce lifestyle-based risk constructs, such as activity risk, interest risk, and opinion risk, and incorporates these into models to evaluate their influence on credit risk. In this regard, we focus on another subjective trait type – emotions. Fortunately, advances in personality extraction from online social media posts inspire us to construct emotions from such textual data, and employ these emotion-related features to evaluate and explain default behavior. Therefore, this study, by taking advantage of online social media, aims to extend the literature on emotions and decision-making by exploring comprehensively the role of emotions in FinTech platform credit risk management.

### 2.2. Emotions and decision-making

A great number of studies have illustrated the association between emotions and decision-making in various contexts [50]. For example, [51] investigate whether consumers' emotions could be leveraged to predict their purchase decision, finding that emotional response is important for making an accurate prediction. [52] study the impact of emotional information on service popularity and sales in space-sharing platforms. They find a positive effect of emotional expressions at low levels, but it diminishes at high levels. [53] focus on online charity, and they show evidence that multi-modal emotional expression plays an important role in online charity crowdfunding success. [54] guide investment decision-making utilizing Twitter sentiment to predict stock returns. [55] explore how anticipated post-purchase regret impacts consumer search behavior and its influence on price and assortment competition among firms. They reveal that anticipated regret drives consumers to conduct more thorough searches among competitive alternatives, resulting in heightened price competition. [56] examines the role of consumer emotions in their online review writing behavior, and finds that consumers tend to comment more in reviews when they have an extreme emotion, either positive or negative, with a higher magnitude for extremely negative emotion than for extremely positive emotion. [57] discover that individuals tend to choose more exciting options when their happiness is focused on the future and more calming options when focused on the present moment. [58] connect weather-induced emotions and consumers' product variety-seeking behaviors. [59], having first examined how consumers' preferences are

affected by the interaction between their arousal level and the valence of their current affective state, show that consumers will tend to choose products that are congruent with their current level of arousal if they are in a pleasant mood, and will choose products that are incongruent with their current level of arousal if they are in an unpleasant mood.

Among all emotions, happiness and anger have shown prevalence and significant impacts in the information systems (IS) contexts. Generally, happiness and anger are both commonly expressed and readily identifiable in individuals' online environments [60,61], making them pragmatic digital footprints for understanding user behavior. For instance, [60] show that happiness in online reviews simplifies cognitive processing, enhancing perceived helpfulness, whereas the impact of anger on review helpfulness varies depending on the context. [62] note that anger's impact on the helpfulness of online reviews varies with the context. [63] find that anger in reviews by females reduces perceived helpfulness due to lower credibility, while male anger does not affect it. [64] observe that anger in medical crowdsourcing platforms increases user engagement, suggesting a pragmatic utility of anger in enhancing interaction. Additionally, [61] find that both happiness and anger can influence judgment biases in algorithmic assessments, with the anchoring effect of AI scores impacting human judgments.

Research on the role of emotions in FinTech platforms is rare, and that which exists focuses primarily on the likelihood of getting a loan approval or project funding. For example, [65] find that borrowers' facial expressions of happiness and sadness can generally increase the amount of funding raised, but when emotional intensity increases, a negative effect on funding occurs. [66] reveal that by expressing negative emotions in the descriptions, borrowers can prolong the funding time and thus improve funding success. A small strand of literature, in particular, leverages soft information with sentiment valence and associates it with credit risks. For example, [67] uncover that the use of positively framed messages is positively associated with positive funding outcomes and enhances the positive impact of the credit ratings on funding outcomes. [68] find that conditional on being funded, text-related factors which reflect emotions hardly predict default probabilities in peer-to-peer lending. Similarly, [69] reveal that borrowers whose writing is more positive are less likely to default on a debt crowd-funding platform. But none of these studies dives into the underlying mechanism of emotions to interpret their findings.

Theoretically, happiness and anger are among the most relevant emotions concerning borrowers' credit behavior due to their broad dimensions and strong action tendencies [70,71]. Compared to other emotions such as sadness or fear—often characterized by passive or avoidant responses—happiness and anger exhibit stronger action tendencies that directly influence repayment decisions [71].

Notably, although individuals may selectively reveal or mask certain emotional states online, growing evidence suggests that social media-expressed emotions often align closely with users' felt emotions. Large-scale analyses show that fluctuations in people's posts tend to mirror offline mood patterns [72], while a large-scale study combining daily mood self-reports with aggregated social media sentiment demonstrated strong correlations between text analysis results and actual levels of self-reported mood, suggesting that emotion dynamics can be accurately traced through online textual content [73]. Moreover, social media platforms often afford some degree of anonymity or pseudonymity [74], which often encourages more candid disclosure of both positive and negative feelings. Although there is potential for exaggeration or distortion, individuals who perceive greater anonymity are generally more inclined to express their true feelings without fear of immediate social repercussions [75]. This can foster more open discussions of personal challenges such as emotional distress and mental health concerns [76]. Hence, while no measure of self-expression is perfect, researchers widely accept that such online posts provide meaningful proxies for users' underlying emotional states.

Our work differs from the above papers in several aspects, showcasing our unique contributions. First, the existing literature, with

similar goals to our study, focuses on general soft information, and uses a single and compound metric to measure emotions from negativity (anger/sadness) to positivity (happiness)—it is closer to the concept of sentiment (e.g., 67–69). Sentiments involve aggregated emotional dispositions or feelings formed toward an object and are enduring. In contrast, emotions concern a broader spectrum and provide greater granularity in capturing affective states. That is, emotions should be identified into different types. These different types of emotions, such as anger and happiness, are sometimes not mutually exclusive [77]. Besides, we investigate the role of emotions in explaining credit risks by considering an additional factor, namely, the temporal distance between emotion measurement (i.e., occurrence) and time to fulfill the financial obligation. Finally, leveraging the potential of social media to monitor borrowers' emotions post-loan, we extend the literature on repayment-reminder strategies by interacting reminder strategies with individual emotions. This could help inform post-loan management with more personalized strategies.

### 3. Theoretical foundation and hypothesis development

#### 3.1. Theoretical foundation

Emotions are complex states comprising subjective experiences, physiological arousal, and behavioral expressions that typically arise when individuals appraise salient events or information in their environment [78]. In this study, we draw on two complementary theories—*appraisal theory of emotions* and *feelings-as-information theory*—to explore whether and how emotions are associated with individuals' credit behavior in the FinTech context. Appraisal theory explains how external events, such as daily occurrences, looming deadlines, or intervention messages, trigger emotional arousal, while feelings-as-information explains how these expressed emotions guide borrowers' loan-related decisions.

Emotions are complex states comprising subjective experiences, physiological arousal, and behavioral expressions that typically arise when individuals appraise salient events or information in their environment [78]. Appraisal theory of emotions indicates that people continuously evaluate how relevant and controllable events are for their goals or well-being [17,18]. Hence, day-to-day life occurrences (e.g., conflicts, achievements) and time-sensitive financial stimuli (e.g., impending deadlines, reminder messages) may evoke positive or negative emotions based on how borrowers perceive their implications [79]. These appraisals manifest across various communication channels, including social media. For instance, individuals who feel threatened by upcoming bills might express anxiety online, whereas borrowers who perceive supportive conditions may share posts reflecting confidence. Appraisal theory has been widely applied to explain individuals' emotional responses in diverse settings, such as clinical psychology [80], organizational behavior [81], and online shopping [22].

Additionally, feelings-as-information theory [19,82] suggests that individuals often rely on their current affective states as heuristics or cues when making judgments under uncertainty. Specifically, individuals use their emotional states as informational cues for judgments, especially when there is no reason to discount these emotions. Emotions act as heuristics, simplifying decision-making across various contexts, from personal well-being to broader social perceptions [83].

#### 3.2. Hypotheses development

Building on the integration of appraisal theory and feelings-as-information theory, we propose that borrowers' emotions, as expressed on social media, are associated with credit risk behaviors in several key ways. Specifically, we develop three primary hypotheses concerning the relationships between borrowers' credit behavior and their expressed emotional intensity, emotional types over time, and emotional interactions with repayment reminders.

### 3.2.1. Emotional intensity and default risks

According to appraisal theory, positive emotions (e.g., happiness) arise when borrowers interpret external events—such as positive feedback or reassuring circumstances—as conducive to their well-being, thus heightening optimism and perceived control [17,18]. At moderate levels, this optimism encourages on-time payments and reduces procrastination by providing balanced affective signals that promote prosocial financial management [84,85]. However, when these emotions intensify, borrowers may develop an unwarranted sense of security, disproportionately skewing their judgment and undermining prudent financial planning—ultimately leading to underestimation of potential risks [23,86].

Likewise, negative emotions such as anger often emerge from perceptions of unfairness or wrongdoing [86,87]. At moderate levels, anger can foster problem-solving or self-advocacy because it triggers a focused “action readiness” without overwhelming logical evaluation [86,87]. According to the feelings-as-information theory, this moderate anger serves as a heuristic signal indicating that something is amiss, prompting closer scrutiny of the situation [19,20]. When anger remains within moderate bounds, it does not distort rational thought or escalate into retaliatory behavior; instead, it provides sufficient arousal to motivate proactive, solution-oriented responses. However, intense anger—particularly if borrowers feel genuinely wronged—may overshadow logical considerations, supporting a retaliatory mindset that rationalizes delayed repayments or even outright defaults [5].

Hence, we posit that borrowers experiencing high or extreme intensities of either positive or negative emotions are more susceptible to default, whether due to overconfidence or retaliatory motives. By contrast, moderate emotional intensities engender more constructive responses and increase the likelihood of responsible debt management. Formally:

**Hypothesis 1.** Borrowers’ high or extreme emotions are positively associated with default likelihood, whereas moderate intensity of emotions is associated with a lower probability of default.

### 3.2.2. Pre-maturity emotional type, temporal distance of repayment, and default risks

Next, we propose that borrowers’ emotions are associated with credit risk behaviors in specific ways that shift over time. The appraisal theory likewise indicates that the meaning and salience of emotional experiences evolve with temporal distance [88,89]. When the repayment deadline is still distant, negative emotions such as anger and dissatisfaction can signal deeper, more persistent concerns about fairness or feasibility [90]. Because these emotions are tied to a borrower’s appraisal of potential injustices or obstacles, they suggest a heightened probability of eventual default. Conversely, positive emotions early in the lending cycle may reflect generalized optimism or a broad sense/feeling of well-being rather than actual preparedness for repayment, rendering it less predictive of final repayment behavior [91].

Meanwhile, although emotions can be influenced by various factors such as salary payments or family emergencies, borrowers may largely experience heightened negative emotions as loan maturity approaches due to increased awareness of impending repayment deadlines [5]. Credgenics [92] indicates that borrowers often experience anxiety and stress regarding their ability to meet financial obligations, especially as repayment deadlines approach. This heightened awareness can trigger feelings of unease and psychological strain, as individuals confront the fear of default and the potential long-term consequences of unresolved debt. As a result, negative emotions may become less diagnostic of actual default behavior [5]. Feelings-as-information theory helps explain why this is the case: once negative affect becomes widespread, it no longer serves as a clear heuristic signal, since virtually all borrowers may feel varying degrees of anxiety or worry. In contrast, remaining positively inclined under looming

pressure stands out as a more exceptional emotional state—offering a stronger indication of confidence and preparedness [20]. Thus, as the deadline approaches, positive emotions provide more distinctive and actionable cues about borrowers’ commitment to repay on time.

Taken together, negative emotions early on can reveal deeper dissatisfaction with personal financial constraints or other personal situations around, indicating a higher likelihood of default. However, once negative affect driven by time pressure becomes intensified, those same emotions tend to lose their predictive value. At that stage, positive emotions emerge as a more informative signal of reliable repayment. This reasoning underpins our hypothesis regarding the interaction between emotional valence, temporal distance, and default behavior. Therefore, we propose:

**Hypothesis 2.** Negative emotions are more likely to be associated with the likelihood of default behavior among borrowers when the repayment deadline is far in the future, whereas positive emotions provide more informative signals when the deadline is closer.

### 3.2.3. Repayment-reminder type, pre-maturity emotional intensity, and repayment behaviors

FinTech platforms frequently use reminder messages (e.g., “positive-expectation” vs. “negative-consequence”) to encourage repayment [16, 44]. We argue that strong emotions—regardless of whether they are positive or negative—amplify borrowers’ responses to repayment reminders at the pre-maturity stage. According to the appraisal theory, emotional intensity leads borrowers to interpret external prompts more intensively [86]. Consequently, when a reminder emphasizes supportive or positive outcomes (e.g., highlighting moral or reputational benefits), these emotionally charged borrowers are more inclined to process this information as reinforcing their feeling of optimism or responsibility, thereby increasing their willingness to repay [16,44].

Conversely, if the reminder spotlights penalties, losses, or negative consequences, those same borrowers may appraise the content as compounding their already heightened sense of threat or perceived unfairness—especially if anger or other negative emotions are involved [45]. Feelings-as-information theory suggests that, intense emotions function as salient internal cues, making borrowers more receptive to the tone of the message [19,20]. Thus, a “negative-consequence” reminder can be interpreted as an additional source of stress or blame, which may deter compliance and even prompt retaliatory or avoidant behaviors. Consequently, under strong emotional intensity at the pre-maturity stage, a “positive-expectation” reminder is more likely to prompt repayment, while a “negative-consequence” reminder risks deterring it. Therefore, we hypothesize:

**Hypothesis 3.** A positive-expectation reminder prompts whereas a negative-expectation reminder deters repayment when platform delinquent borrowers maintain strong emotional intensity at the pre-maturity stage.

Notably, researchers have the flexibility to measure social-media-based emotions over various time periods. We posit that within a relatively short window (e.g., two to three months), emotions measured across different time spans should generally demonstrate similar relationships with borrowers’ credit behaviors. However, long-term measurements may provide more moderate appraisals, as they smooth out short-term fluctuations and offer a more stable representation of emotional tendencies, reducing the impact of transient events. As a result, we anticipate that long-term emotions will align more closely with deliberate financial behaviors [18]. In this study, we examine both short-term and relatively long-term measurements of emotions.

## 4. Context and data

### 4.1. Research context

We collaborated with a leading Asian FinTech platform to collect our data set. The platform, founded in 2011, is funded by its shareholders and runs based on a business-to-consumer model. The platform offers microloans of an average size of approximately USD 450 and had an accumulated transaction volume of more than 3 million USD by the end of 2018. Most loans have a period between one to ten months, and the yearly interest rate ranges from 10 to 16%. To apply for a loan, borrowers need to provide basic personal details such as age, gender, educational background, and income level. The platform collected borrowers' social media activities from a popular Twitter-like microblog website in that country.<sup>6</sup> Thankfully and fortunately, during the sample period of our study, the platform did not employ social media information to assess borrowers' credit risk. This mitigates to a large extent the selection bias of social media usage from the platform's borrower screening process.

Borrowers can access a detailed description of the default process via a module titled "Consequences of Default" on the first page of the platform's official website. If a loan is not fully paid off more than 90 days after the due date, the default behavior is confirmed. In cases of default, the platform takes legal action. The final default records are submitted to a centralized shared blacklist system maintained by a consortium of microloan FinTech platforms. The language used to describe the default process is neutral and not intended to impose an intense sense of threat.

### 4.2. Data and variables

To identify the emotional trends, we focus on borrowers who were continuously using the *microblog* throughout the loan process covering both the pre- and post-loan-application periods. The microblog is the largest online public Twitter-like social media in the focal country. The average word count of a blog is approximately 21. Approximately 1/3 of borrowers in the platform were using the microblog throughout our observation window. The platform offered us 7,000 randomly-selected eligible borrowers of social media usage who applied and were granted loans between January 2015 and December 2016. None of them borrowed money more than once from the focal platform during the above period. After removing records with missing values or borrowers whose total number of blog posts exceeds two standard deviations above the average, our final sample covers 6,826 borrowers. Our observation window is between October 2014 and June 2018, covering at least 3 months prior to loan applications and 3 months after loan (maturity) due dates in our sample. This time window is sufficiently long to observe our sample borrowers' microblog usage in the pre-approval stage (i.e., before submitting the loan application<sup>7</sup>) and confirm their

default behavior. Our data set contains high-resolution borrower and loan information with the following diverse categories of variables.

*Loan attributes:* including loan amount, loan term, yearly interest rate, and borrower's loan purpose and loan history.<sup>8</sup>

*Borrower characteristics:* namely borrowers' demographics and socioeconomic status, including age, gender, education level (i.e., the highest degree a borrower has obtained), monthly income level, number of children, marital status, and housing condition.

*Repayment information:* including a label indicating whether a borrower defaulted (i.e., 1 for default and 0 for non-default) and the days between the paid-off date and loan maturity/due date (i.e., smaller than 0 for payments in advance and larger than 0 for late payments).

*Borrower microblog presence:* including the total number of fans, followings, and blogs posted, and the total number of reposts, comments, and "likes" received; we also code the maximum number of reposts, comments, and "likes" per blog a borrower received, respectively. We count the numbers of fans and followings based on microblog information accrued by June 2018, the end of our observation period. As for the remaining post-related features, we consider posts from the specific borrower within the three months preceding loan approval up to the loan maturity date.

Table 1 summarizes the descriptive statistics for all of the key variables. Although our initial sample covers 6,826 observations, in order to depict borrowers' emotions, we first rely merely on borrowers who posted at least one blog within one month prior to approval. We describe in detail the construction process in Section 4.3. This results in a smaller sample of 1,943 borrowers. The focal borrowers' default rate is 23%, similar to that of other platforms in the market within a similar observation period (e.g., 94). The average age of borrowers is around 23, and males make up 74% of borrowers. Besides, most borrowers have a polytechnic or undergraduate education background, and only 15.6% of them own their own houses.

It is worth noting that the FinTech platform assisted us in comparing key characteristics (e.g., gender, age, monthly income level, loan purpose, and loan amount) of social media borrowers (our study's focal sample) with two other groups: (1) borrowers who used microblog but did not post any messages within one month prior to approval, and (2) borrowers who never used microblog. The statistics revealed no significant differences among these groups, indicating that social media usage (or lack thereof) had minimal impact on borrowers' loan application behavior and affirming the representativeness of our sample.<sup>9</sup>

### 4.3. Emotion extraction

Inspired by [95], for each borrower, we extract, based on his/her microblog textual posts, six basic and common types of emotions, namely anger, happiness, surprise, disgust, fear, and sadness. We ap-

<sup>6</sup> The focal social media website is publicly accessible, and the collection of such information is widely accepted as a standard practice in the focal credit market [93]. In their loan application, we notified borrowers of the potential use of their online social media posts during the evaluation periods and repayment periods, and, correspondingly, asked for their permission to access their accounts. Given that the platform is publicly accessible and borrowers are typically in urgent financial need, the platform manager informed us that no borrower during our data period declined to authorize data access, except in cases where they did not have a social media account. Additionally, borrowers were not informed about the specific social media information that would be used or which posts or signals might influence their loan approval probability. They were also unaware of the exact timing of the social media data collection. This process likely minimized borrowers' ability to strategically modify their social media posts after submitting their applications, ensuring that the data reflects a comprehensive and authentic snapshot of their online activity.

<sup>7</sup> The focal platform screened loan applications in a "first-come-first-served" manner and typically made decisions within one day of receiving applications.

<sup>8</sup> Although borrowers had multiple options in choosing loan amounts and loan terms, most loan products concentrated on 5, 6, and 7 months (accounting for more than 85% of the observations in our sample) with approximately USD 450. Unlike other microloan platforms, the interest rates on the focal platform are exogenously determined on a daily basis, and are orthogonal to individuals' credit scores, loan terms, or amounts.

<sup>9</sup> We also conduct topic modeling using BERTopic on borrowers' social media posts prior to loan approval and prior to loan maturity (conditional on delinquency). The results reveal diverse and overlapping topics across both stages, suggesting minimal strategic posting and supporting the representativeness of the social media samples. Additional details are available upon request.

**Table 1**  
Descriptive statistics.

| N=1943                                             | (1)     | (2)      | (3)  | (4) |
|----------------------------------------------------|---------|----------|------|-----|
|                                                    | Mean    | Std. Dev | Max  | Min |
| <i>Repayment performance</i>                       |         |          |      |     |
| Default                                            | 0.230   | 0.421    | 1    | 0   |
| <i>Emotions at the pre-approval stage</i>          |         |          |      |     |
| Immediate anger                                    | 0.150   | 0.152    | 1    | 0   |
| Average anger tendency                             | 0.152   | 0.127    | 1    | 0   |
| Immediate happiness                                | 0.238   | 0.160    | 1    | 0   |
| Average happiness tendency                         | 0.243   | 0.135    | 1    | 0   |
| <i>Borrower characteristics</i>                    |         |          |      |     |
| Monthly income level                               | 2.412   | 1.405    | 9    | 1   |
| Education level                                    | 2.576   | 0.505    | 4    | 1   |
| Gender (1 = male)                                  | 0.743   | 0.437    | 1    | 0   |
| Age                                                | 22.893  | 1.155    | 37   | 21  |
| #Children                                          | 0.006   | 0.132    | 5    | 0   |
| Marital status (1 = in a marriage)                 | 0.012   | 0.108    | 1    | 0   |
| House ownership (1 = self-own)                     | 0.156   | 0.363    | 1    | 0   |
| <i>Loan attributes</i>                             |         |          |      |     |
| Loan amount (USD)                                  | 454.914 | 43.475   | 465  | 155 |
| Loan purpose (1 = used for emergency)              | 0.376   | 0.484    | 1    | 0   |
| Loan term (month)                                  | 6.557   | 2.549    | 25   | 1   |
| Loan history (1 = borrowed on the platform before) | 0.155   | 0.362    | 1    | 0   |
| Yearly interest rate (%)                           | 14.103  | 0.995    | 16   | 10  |
| <i>Microblog presence (static)</i>                 |         |          |      |     |
| Total #blogs posted                                | 205.952 | 299.234  | 1162 | 6   |
| Total #fans                                        | 220.232 | 392.121  | 1572 | 9   |
| Total #followings                                  | 207.729 | 171.567  | 663  | 28  |
| Total #reposts received                            | 12.907  | 29.194   | 115  | 0   |
| Total #comments received                           | 74.733  | 145.923  | 568  | 0   |
| Total #“likes” received                            | 91.826  | 169.600  | 659  | 0   |
| Max. #reposts a blog received                      | 1.881   | 2.723    | 9    | 0   |
| Max. #comments a blog received                     | 4.073   | 3.364    | 9    | 0   |
| Max. #“likes” a blog received                      | 3.885   | 2.980    | 9    | 0   |

<sup>a</sup> Since we consider only a subsample for analyzing the effects of emotions at the pre-maturity stage, we do not report the stats of emotions at the pre-maturity stage here. Refer to Section 5.3 and Table A.2 in the appendix for more details.

<sup>b</sup> Monthly income level: 1 = USD300 or below, 2 = USD300–USD600, 3 = USD600–USD900, ..., 8 = USD2100–USD2400, 9 = USD2400 or above. Education level: 1 = high school or below, 2 = polytechnics, 3 = undergraduate, 4 = postgraduate.

ply a popular dictionary, *SC-LIWC*, to extract emotions.<sup>10</sup> *SC-LIWC* is an extended version of the *LIWC* (Linguistic Inquiry and Word Count) dictionary, specifically designed for language analysis in social media (i.e., social computing). It evaluates emotions, social interactions, cognitive processes, and psychological states expressed in writing [96]. Specifically, we utilized the Chinese version of *SC-LIWC* (<https://cliwcg.weebly.com>).

As we have discussed above, borrowers' emotions are influenced by external circumstances and might alter frequently. Thus, we propose to identify two kinds of emotion trends: immediate emotion (i.e., an immediate psychological status that occurs at a particular time point) and average emotion tendency (i.e., an average psychological status over a long term). Empirically, we use microblogs posted within one month prior to loan approval or within one month prior to loan maturity to measure immediate emotions at the pre-approval stage or the pre-maturity stage, respectively. By contrast, to measure average emotional tendencies at the pre-approval stage and at the pre-maturity stage, we use microblogs posted within three months prior to loan approval and within three months prior to loan maturity, respectively. Considering that sentiment analysis, in general, cannot be performed precisely with very short texts such as microblogs or tweets due to low information density [97], we follow a standard sentiment-mining operation to extract emotions.

<sup>10</sup> We also conduct a robustness check by leveraging a large language model (LLM)—specifically, the Azure OpenAI GPT-35-turbo-16k model—to extract emotion scores in Chinese. The prompt used is “Please extract emotion scores based on the following text content. The scoring range is from 0 to 1 for each emotion”. All empirical results remain robust. More discussions are offered in Section 5.4.

Specifically, for each microblog post, we sequentially conduct tokenization and sentence segmentation, remove stop words using the stop-word list provided by *SC-LIWC*, and derive emotion scores within the range of 0 to 1 by matching the sample texts to the *SC-LIWC* dictionary. We finally average each borrower's microblog posts within one and three months to obtain immediate emotions and average emotional tendency. Fig. 1 presents the timeline for granted loans and the observation period for borrowers' social media posts at the pre-approval and pre-maturity stages for emotion score calculation.

Table A.1 presents the correlation matrices among default, immediate emotions, and average emotional tendencies, respectively. There are some interesting implications. First, we do not find very high correlations of the same emotion type's presence between the pre-approval and pre-maturity stages (see the shaded cells), except for sadness. For example, the correlation between immediate anger at the pre-approval and pre-maturity stages is merely 0.084. This suggests that borrowers' emotions might dramatically change with time; so, it should be considered plausible to investigate separately the role of emotions at both the pre-approval and pre-maturity stages in predicting borrowers' credit risks.

Second, there are two higher-level emotion categories: happiness and surprise (positive emotions), and, anger, disgust, fear, and sadness (negative emotions) [98]. Our statistics in Table A.1 show that pairwise correlations between any two specific emotion types within the positive/negative emotion categories are very high (e.g., 0.837 between happiness and surprise, 0.790 between disgust and sadness, and 0.539 between anger and sadness at the approval stage in Panel A; 0.867 between happiness and surprise, 0.808 between disgust and sadness, and 0.638 between anger and sadness at the approval stage in Panel B).<sup>11</sup> Moreover, both happiness and anger demonstrate higher levels

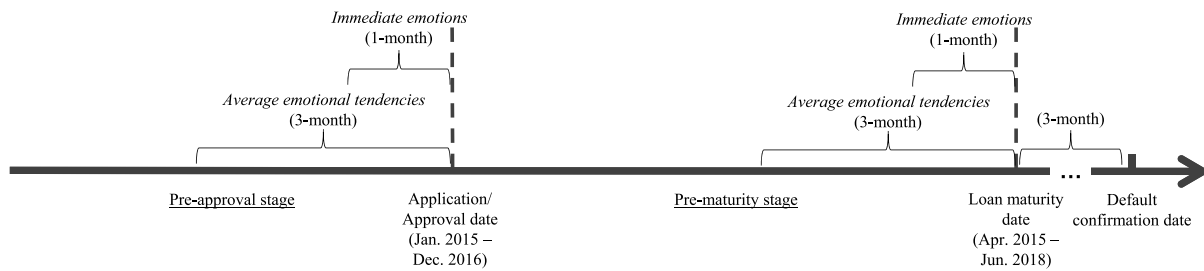


Fig. 1. Timeline for granted loans and social media posts' observation period.

and high variances than other emotion types; for example, immediate happiness has a mean of 0.238 with a standard deviation (SD) of 0.160, while surprise has a mean of 0.169 with an SD of 0.140. Immediate anger has a mean of 0.150 with a standard deviation (SD) of 0.152, while other negative emotions such as fear, have a mean of 0.075 with an SD of 0.098, and sadness has a mean of 0.126 with an SD of 0.099, and these differences hold true for average emotional tendencies (results available upon requests). The high variance helps to capture more diverse/variant positive and negative emotions among borrowers, which paves the way to check the potential association between emotions and repayment behavior with such more representative emotions. Therefore, in our main analyses, we majorly consider the effects of happiness and anger.

Third, interestingly, we find that the correlations between happiness and anger are positive and large regardless of the measurements or occurrence stages (i.e., 0.434, 0.394, 0.462, and 0.416 in Table A.1). This implies that positive and negative emotions present qualitatively different dimensions, and thus they may present both high and low at the same time in a person [99]. This empirical evidence consolidates the necessity of investigating emotions rather than a compound sentiment.

## 5. Relationship between emotions and credit risks

In this section, we explore whether borrowers' emotions are correlated with their loan repayment performance. As discussed earlier, we focus on the impact of happiness and anger in our main analyses. Unlike other classic characteristics such as demographic features or alternative information like shopping activities, emotions could be unstable, especially given changes to the external environment. Hence, we run two disparate empirical analyses for credit risk assessment at the loan (pre-)approval (i.e., for the platform's loan-approval decisions) and pre-maturity stages.

### 5.1. Addressing empirical challenges

It is crucial to clarify that we aim to highlight the association between emotions and FinTech platform credit behavior, which assists FinTech platform practitioners in utilizing social media emotions for default behavior pre-diagnosis, avoiding a causal investigation but revealing a potentially significant connection. While we acknowledge this, we are dedicated to meticulously controlling for other variables that might potentially confound this relationship. We recognize the need for a cautious interpretation of the estimated coefficients, but we have diligently eliminated confounding variables to mitigate the risk of a superficial correlation. This ensures that the coefficient's

direction holds meaningful insight into whether ex-ante emotions play a substantial role in credit behavior. We illustrate next how controlling for these confounding factors enables us to discern the true relationship between emotions and default, ruling out possible spurious correlation or endogeneity issues.

Borrower characteristics, such as income and education level, gender, age, marital status, whether having children, and whether owning a house, can be correlated with both probability of default and emotions, respectively. For example, borrowers with higher incomes are more likely to have the financial means to make loan payments and thus have lower default rate [100]; at the same time, they may experience lower levels of stress and anxiety related to loan repayment, as they have a greater ability to meet their financial obligations.

Loan attributes, such as loan amount, loan purpose, loan term, and loan interest rate can play a very important role in both credit default and emotions. For instance, the burden of repayment increases as the loan amount grows, and the approval of this loan can also be correlated with a heightened sense of delight upon the loan's approval.

Another concern is that the usage of microblogs itself may contain emotional information from borrowers. Borrowers who do not use microblogs frequently may not post extensively, which could result in the information within their posts not being informative enough to accurately characterize their emotions. To address this selection issue, we control for borrowers' microblog presence, such as their total number of posts, fans, followings, comments, and reposts.

We also control for borrowers' living province fixed effects and loan application month fixed effects, for the following reasons. Firstly, different regions in the focal country could possess different macroeconomic environments that might influence the accessibility of loan channels for borrowers. Additionally, the timing of loan applications might correlate with both emotions and future loan repayment. Microloans are time-sensitive transactions, susceptible to the impact of evolving regulatory policies affecting borrowers' willingness to borrow, repay, and their emotional state. Furthermore, weather conditions during different seasons could also exhibit a correlation with borrowers' overall emotional well-being.

### 5.2. Emotions and credit risks at loan pre-approval stage

We first investigate how borrowers' emotions occurring at the pre-approval stage (i.e., before application) are relevant to their default behavior for microloans. This analysis offers platform practitioners a number of features of emotions that can potentially be used to determine if they should grant loans to certain borrowers.

We conduct empirical analyses at the borrower level. We employ a linear probability model (LPM) to examine the relationships between borrowers' emotions and their default probabilities at the pre-approval stage. We choose LPM for our main analysis, because it is more interpretable than the Logit or Probit models, as its coefficients can be interpreted directly as marginal effects. Recently, LPM has been widely used in the information systems and marketing fields [101,102]. We redo our analyses with logit models and yield consistent results. We specify our model in Eq. (1). The dependent variable Default,

<sup>11</sup> Although some of Ekman emotions such as anger, happiness, and sadness might be theoretically orthogonal, social-media-based emotion measurements often blend due to the aggregation of multiple posts over time, resulting in emotions appearing more interrelated than traditional discrete emotion models would predict.

**Table 2**  
Emotions at the pre-approval stage and loan default.

| DV: Default               | Immediate emotions   |                      |                      |                      | Average emotional tendencies |                      |                      |                      |
|---------------------------|----------------------|----------------------|----------------------|----------------------|------------------------------|----------------------|----------------------|----------------------|
|                           | Anger                |                      | Happiness            |                      | Anger                        |                      | Happiness            |                      |
|                           | (1)                  | (2)                  | (3)                  | (4)                  | (5)                          | (6)                  | (7)                  | (8)                  |
| Emotion_a                 | −0.047<br>(0.063)    | −0.392***<br>(0.136) | −0.051<br>(0.060)    | 0.202<br>(0.157)     | −0.045<br>(0.076)            | −0.506***<br>(0.158) | −0.025<br>(0.072)    | 0.040<br>(0.192)     |
| Emotion_a_square          |                      | 0.593***<br>(0.206)  |                      | −0.391<br>(0.244)    |                              | 0.815***<br>(0.245)  |                      | −0.103<br>(0.282)    |
| Constant                  | −0.983***<br>(0.303) | −0.923***<br>(0.303) | −0.980***<br>(0.303) | −1.018***<br>(0.304) | −0.987***<br>(0.303)         | −0.920***<br>(0.303) | −0.990***<br>(0.303) | −0.998***<br>(0.304) |
| Controls                  | Yes                  | Yes                  | Yes                  | Yes                  | Yes                          | Yes                  | Yes                  | Yes                  |
| Loan application month FE | Yes                  | Yes                  | Yes                  | Yes                  | Yes                          | Yes                  | Yes                  | Yes                  |
| Living province FE        | Yes                  | Yes                  | Yes                  | Yes                  | Yes                          | Yes                  | Yes                  | Yes                  |
| Mean of Emotion_a         | 0.150                | 0.150                | 0.238                | 0.238                | 0.152                        | 0.152                | 0.243                | 0.243                |
| Adj. R-squared            | 0.049                | 0.053                | 0.049                | 0.050                | 0.049                        | 0.055                | 0.049                | 0.049                |
| N                         | 1943                 | 1943                 | 1943                 | 1943                 | 1943                         | 1943                 | 1943                 | 1943                 |

<sup>a</sup> Models (1), (2), (5), and (6) correspond to models using anger as independent variables in Emotion\_a and Emotion\_a\_square; Models (3), (4), (7), and (8) use happiness. Suffix “a” stands for emotions at the pre-approval stage.

<sup>b</sup> Standard errors are in parentheses. \*\*\*, \*\* and \* represent statistical significance at the 1%, 5% and 10% levels, respectively.

<sup>c</sup> Control variables include borrower characteristics, loan attributes, and microblog presence. Refer to Table B.1 for full regression results.

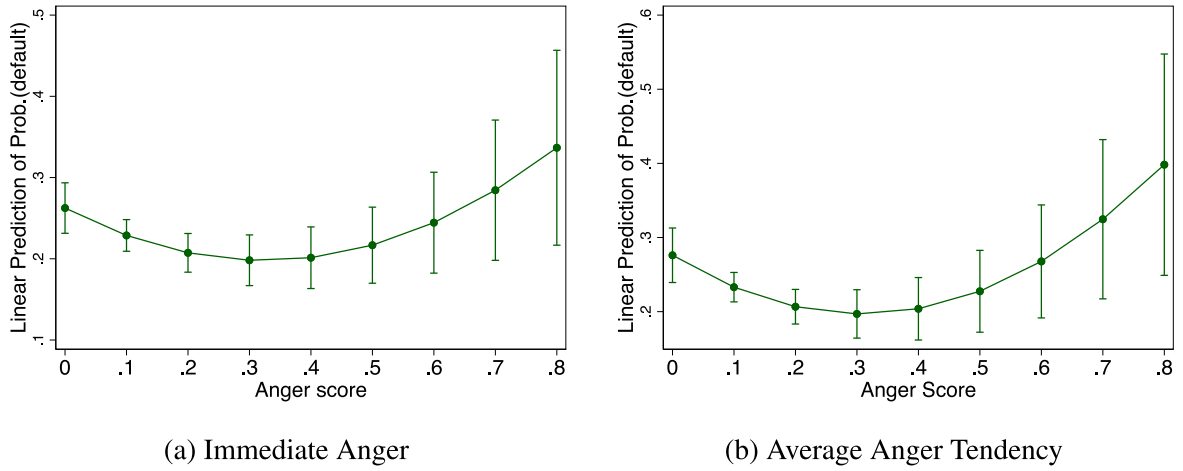


Fig. 2. Predictive margins of anger at the pre-approval stage on default (with 95% CIs)

is a dummy variable indicating if borrower  $i$  failed to fully repay his/her loan ( $\text{Default}_i = 1$  for default, 0 for non-default, i.e., repaid in advance, on time, or became delinquent-but-not-in-default). Notably, as discussed in Section 2.2, different intensities of emotion may exert opposite relationships on decision-making; for example, extremely negative emotions would correlate to socially deviant behavior, whereas the moderate intensity of negative emotions may invoke prosocial behavior [23]. Therefore, we include a linear term  $\text{Emotions\_a}_i$  and a quadratic term  $\text{Emotions\_a\_square}_i$  for the focal emotions (i.e., happiness or anger) in the model in order to capture the nonlinear influential patterns of emotions on default. That is,  $\text{Emotion\_a}$  is applied to Anger or Happiness in the model estimations. The suffix “a” stands for emotions at the pre-approval stage.

$$\text{Default}_i = \beta_0 + \beta_1 \cdot \text{Emotions\_a}_i + \beta_2 \cdot \text{Emotions\_a\_square}_i + \vec{X}_i \gamma + \epsilon_i. \quad (1)$$

We have eight models, corresponding to any of the pairwise combinations of the two emotion types (i.e., Anger and Happiness), two measurements (i.e., immediate emotions and average emotional tendencies), and two model specifications (i.e., only including the linear term and additionally including the quadratic term). We control for factors that may be associated with default behavior as well, and we denote them as  $\vec{X}_i$ . Specifically, we consider controls, as listed in Table 1, covering borrowers’ characteristics, loan attributes, and microblog presence. As indicated by Lusardi and Mitchell [103], borrowers’ education may have a profound relationship with their background characteristics such as monthly income, which in turn associates with

credit behavior. Guided by this insight, we incorporate the interaction effects of education level and monthly income into our regression analyses. Moreover, as borrowers’ repayment behavior might present temporal and regional discrepancies [104], we also include fixed effects of the loan application/approval time (months) and borrowers’ living provinces.

### 5.2.1. Empirical results

Models (1) to (4) in Table 2 report results with immediate emotions, and models (5) to (8) report regression results with average emotional tendencies. Models (1), (2), (5), and (6) show regression results when using anger as the independent variable in  $\text{Emotion\_a}$  and  $\text{Emotion\_a\_square}$ , and models (3), (4), (7), and (8) show regression results when using happiness. The results show that happiness at the pre-approval stage has an insignificant correlation with loan defaulting, regardless of immediate happiness or average happiness tendency. A possible explanation is that the role of happiness in self-control (from financial deviation) is not straightforward but hinges on multiple factors such as contextual stimuli and personal goal setting [105].

With respect to anger, interestingly, the significant estimates of  $\text{Emotion\_a\_square}$  demonstrate U-shaped relationships between anger and loan defaulting, in the cases of both immediate anger (Model (2)) and average anger tendency (Model (6)). The axis of symmetry is located at approximately  $\text{Anger\_a} = 0.3$  (i.e., around one standard deviation from the mean value). Figs. 2(a) and 2(b) plot the predictive margins of  $\text{Default}_i$  at different anger (immediate anger and average

**Table 3**  
Emotions at the pre-maturity stage and loan default.

| DV: Default Delinquency = 1   | Immediate emotions |                   |                   |                   | Average emotional tendencies |                   |                   |                    |
|-------------------------------|--------------------|-------------------|-------------------|-------------------|------------------------------|-------------------|-------------------|--------------------|
|                               | Anger              |                   | Happiness         |                   | Anger                        |                   | Happiness         |                    |
|                               | (1)                | (2)               | (3)               | (4)               | (5)                          | (6)               | (7)               | (8)                |
| Emotion_m                     | -0.099<br>(0.165)  | -0.181<br>(0.365) | 0.139<br>(0.195)  | 0.166<br>(0.430)  | -0.142<br>(0.185)            | -0.691<br>(0.427) | 0.195<br>(0.172)  | -0.856*<br>(0.459) |
| Emotion_m_square              |                    | 0.147<br>(0.580)  |                   | -0.048<br>(0.674) |                              | 0.919<br>(0.625)  |                   | 1.487**<br>(0.602) |
| Constant                      | -1.097<br>(0.875)  | -1.082<br>(0.878) | -1.109<br>(0.875) | -1.118<br>(0.885) | -1.015<br>(0.877)            | -0.918<br>(0.878) | -1.085<br>(0.877) | -0.874<br>(0.876)  |
| Controls                      | Yes                | Yes               | Yes               | Yes               | Yes                          | Yes               | Yes               | Yes                |
| Loan application month FE     | Yes                | Yes               | Yes               | Yes               | Yes                          | Yes               | Yes               | Yes                |
| Borrowers' living province FE | Yes                | Yes               | Yes               | Yes               | Yes                          | Yes               | Yes               | Yes                |
| Mean of Emotion_m             | 0.160              | 0.160             | 0.180             | 0.180             | 0.199                        | 0.199             | 0.281             | 0.281              |
| Adj. R-squared                | 0.118              | 0.116             | 0.119             | 0.116             | 0.113                        | 0.116             | 0.115             | 0.127              |
| N                             | 437                | 437               | 437               | 437               | 437                          | 437               | 437               | 437                |

<sup>a</sup> Models (1), (2), (5), and (6) correspond to using anger as independent variables in Emotion\_m and Emotion\_m\_square; Models (3), (4), (7), and (8) correspond to happiness. Suffix “m” stands for emotions at the pre-maturity stage.

<sup>b</sup> Standard errors are in parentheses. \*\*\*, \*\* and \* represent statistical significance at the 1%, 5% and 10% levels, respectively.

<sup>c</sup> Control variables include borrower characteristics, loan attributes, and microblog presence. Refer to Table B.2 for full regression results.

anger tendency, respectively) levels at the pre-approval stage.<sup>12</sup> We focus on the marginal effects. For borrowers who have an immediate angry emotion lower than 0.3 (0.331 exactly, and  $0.331 = -(-0.392/(2 \times 0.593))$ ), the marginal effects of anger on default are negative; for example, for the anger scored of 0.1, a 10-percentage-point-higher anger intensity (score) is on average accompanied by a lower default probability of 2.73 percentage points ( $= |0.1 \times (2 \times 0.593 \times 0.1 - 0.392)|$ ). In contrast, when the immediate angry emotion is higher than 0.331, the marginal effects of anger on default are positive; for anger scored at 0.5, a 10-percentage-point-higher anger would correlate with an increased default probability of 2.01 percentage points ( $= |0.1 \times (2 \times 0.593 \times 0.5 - 0.392)|$ ). Similar patterns are found for average anger tendency. These findings are in line with the literature (e.g., 106) that determines that negative emotions may stem from peoples' cognitive appraisals of uncertain situations; thus generally, people with negative emotions tend to behave carefully, pay attention to external circumstances, and show moral behavior as a consequence. In our case, borrowers who are in an angry mood and experiencing financial difficulty specifically attend to controlling their financial situations and, thus, preventing themselves from default. Note that, 1,753 out of 1,943 (i.e., 90.2%) borrowers in our sample have an immediate anger level lower than 0.331 at the pre-approval stage (93.9% for the average anger tendency case); that is, only a very few borrowers have an “extremely” high intensity of anger before their loan application. Our results suggest that they would be more likely to default on their loans, which suggests that only extremely strong negative emotions would correlate to financially deviant behavior.

### 5.3. Emotions and credit risks at loan pre-maturity stage

Online public social media allows FinTech platform practitioners to track borrowers' emotional statuses after loan approvals; therefore, in addition to leveraging borrowers' emotions for credit risk evaluation and loan-approval decision-making, we also focus on the value of emotions in the post-approval period. An important touch point between the microloan platform and borrowers is the period around the loan maturity dates. The repayment stress as the maturity date approaches may influence borrowers' emotional states. Understanding whether emotions and other variables are related to borrowers' eventual default behavior helps the platform design appropriate strategies to improve paid-off ratios. Therefore, in this subsection, we focus on only the sample of borrowers who did not fully repay their debts by

the end of the loan maturity dates, namely, delinquent borrowers.<sup>13</sup> Also, we investigate whether these borrowers' recent emotions prior to the loan maturity dates (i.e., at the pre-maturity stage) are useful for evaluating their default behavior. Notice that similar to analyses of the pre-approval stage, in order to construct emotional features at the pre-maturity stage, we have to further rely on borrowers who posted at least once within one month prior to the loan maturity dates. We further remove samples who paid off the loans in advance or on time. This results in a sample of 437 delinquent borrowers. We report the descriptive statistics on these 437 borrowers in Table A.2.

Consistent with Eq. (1), we estimate with LPM and include a linear term Emotion\_m and a quadratic term Emotion\_m\_square of happiness and anger in Eq. (2); suffix “m” denotes emotions at the pre-maturity stage. Note that, since we take only the delinquent borrowers into account, we aim to explore the probability of eventual default conditional on delinquency (i.e.,  $\text{Default}_i | \text{Delinquency}_i = 1$ ). Following Eq. (1), we also incorporate borrowers' characteristics, loan attributes, microblog presence, and two sets of fixed effects as control variables (i.e.,  $\vec{X}_i$ ) in Eq. (2).

$$\text{Default}_i | (\text{Delinquency}_i = 1) = \beta_0 + \beta_1 \cdot \text{Emotions\_m}_i + \beta_2 \cdot \text{Emotions\_m\_square}_i + \vec{X}_i \gamma + \epsilon_i. \quad (2)$$

#### 5.3.1. Empirical results

Table 3 reports the estimation results. To justify the nonlinear effects of emotions on default at the pre-maturity stage, we also specify a linear form for comparison. Models (1) to (4) in Table 3 report the regression results with immediate emotions, and Models (5) to (8) report the regression results with average emotional tendencies. Models (1), (2), (5), and (6) show the regression results when using anger as the independent variable in Emotion\_m and Emotion\_m\_square, and Models (3), (4), (7), and (8) show the regression results when using happiness.

Differing from the results with respect to emotions at the pre-approval stage and loan defaulting, for platform delinquent borrowers, their emotions at the pre-maturity stage present overall-trivial relationships with their eventual default behavior. Specifically, we uncover that the nonlinear relationship between anger at the pre-approval stage and loan defaulting but anger at the pre-maturity stage is not useful for

<sup>12</sup> Predictive margins predict  $\text{Default}_i$  at different anger levels, leaving all other covariates the same.

<sup>13</sup> Our focal FinTech platform defines delinquency strictly by the loan maturity date, while we acknowledge that other financial practices may sometimes incorporate grace periods. Regardless of the definition used, our goal is to assess whether social media-expressed emotions can predict borrowers' failure to repay by maturity. This offers a practical tool for timely debt collection planning and can be readily adapted to platforms that include grace periods in their delinquency definitions.

predicting platform delinquent borrowers' loan defaulting, regardless of immediate anger or average anger tendency. This might be due to the fact that delinquent borrowers who expect their potential delinquency would be generally stuck in negative emotional statuses (for their failure to repay on time or other reasons) [5];<sup>14</sup> negative emotions are thus less predictable of the eventual default. However, we find that the average happiness tendency of delinquent borrowers around the loan maturity period presents a U-shaped relationship with default. That is, the effects of average happiness tendency decrease initially but reverse at the score of 0.288 ( $= -(-0.856/(2 \times 1.487))$ ). The mean value of average happiness tendency at the pre-maturity stage is 0.281 in our sample, which is close to 0.288, suggesting that happiness at the pre-maturity stage is contingent on loan default with delinquent borrowers. Particularly, as our argument in the hypothesis development, if a borrower is keeping a relatively high intensity of happiness in the long run, which by intuition, goes against a "normal" mental response of worry or anger to potential delinquency in the near future, he/she tends to default. A plausible explanation is that this kind of borrowers may have determined to default to ease themselves from the financial obligation. By contrast, borrowers with a moderate intensity of happiness are more likely finally to repay their debts. Their relatively favorable emotions in the long term suggest that they maintain a sound mental status about their debts [107] and that delinquency may occur simply due to missing the due dates.<sup>15</sup>

Combining Sections 5.2 and 5.3, the significant results in Columns (2) and (6) of Table 2 and Column (8) of Table 3 support Hypothesis 1. Furthermore, the significant results in Columns (2) and (6) of Table 2, coupled with the insignificant results in Columns (2) and (6) of Table 3, suggest that negative emotions are more strongly associated with the likelihood of default behavior among borrowers when the repayment deadline is far in the future (i.e., when the gap between the pre-approval stage and the loan deadline is large). Additionally, the significant results in Column (8) of Table 3, alongside the insignificant results in Columns (4) and (8) of Table 2, indicate that positive emotions provide more informative signals when the deadline is near (i.e., when the gap between the pre-maturity stage and the loan deadline is small), thereby supporting Hypothesis 2.

#### 5.4. Robustness checks

We run four additional analyses for robustness checks. First, considering that our dependent variable is binary, we redo our analyses in Sections 5.2 and 5.3 with logit models and report consistent results. Second, we incorporate anger and happiness into one regression instead of running them separately, for both immediate emotions and average emotional tendencies at the pre-approval and pre-maturity stages. Third, we examine the potential impact of different time frame selections. As a result, we prolong the measurement of average emotion tendency from three months to six months at both the pre-approval and pre-maturity stages for happiness and anger, and replicate the empirical regressions with LPM. Lastly, we replicate all the main analyses with the emotion scores we output by the GPT-35-turbo-16k model. LLM usage can validate our SC-LIWC-based results and/or to alleviate the potential concern about possible methodological biases. The results are reported in Appendix Tables B.4 and B.5. All above robustness testing results show consistent patterns with those in Tables 2 and 3, reflecting

<sup>14</sup> We are able to determine that the intensities (scores) of anger at the pre-maturity stage for platform delinquent borrowers are much higher than those for non-delinquent borrowers.

<sup>15</sup> The average number of overdue days for the delinquent borrowers who maintain a moderate average happiness tendency at the pre-maturity stage is evidently smaller than those who maintain an extreme average happiness tendency, implying that these borrowers became delinquent not on purpose but rather because they missed due dates.

the robustness of our findings (Robustness check results available upon requests).

Note that although we have discussed the conceptual disparity between emotions and sentiment (refer to Section 2.2), we also extract a variable *sentiment* from the microblogs using classical sentiment mining [38]. We do not find a significant association of sentiment with borrowers' credit risks. This finding, again, offers empirical support for the necessity of differentiating sentiment and emotions.

## 6. Responses with different emotions to repayment reminders

In this section, we investigate the causal relationship between repayment reminder and credit default, moderated by borrowers' emotions. Our secondary-data-based analyses show that borrowers' emotions at the pre-approval stage are highly correlated with their future repayment performance, and that the FinTech platform can leverage this information to enhance its credit risk evaluation and loan-approval decisions. At the pre-maturity stage, however, emotions could not, in general, be helpful for identifying high-risk borrowers from low-risk ones, in spite of having emotion information in hand. Does this mean that the platform could do nothing with borrowers' emotions at the pre-maturity stage? Recall, from Section 2.2, that the literature has implied that with different emotions, borrowers have heterogeneous responses to external interventions [108]; therefore, we run a field experiment on the focal platform to investigate whether and how borrowers with different emotions might respond to repayment-reminder strategies around the loan maturity period. The experiment with randomized assignment and exogenous treatment allows us to identify the causal effects of reminder intervention strategies on borrowers with diverse emotional statuses.

### 6.1. Experimental design and setup

Sending borrowers repayment-reminder messages via SMS or mobile apps at around loan maturity dates is a common practice. Inspired by [16], we focus on two types of repayment-reminder intervention strategy: a message delivering a platform's (lenders') positive expectations about repayment, and a message deterring with negative consequences of default. The positive-expectation message would in general encourage prosocial compliance [16], and a negative-consequence message is more likely to cause anxiety, embarrassment, and shame [44]. [108] suggest that people may reap emotional benefits from communications conveying positive supports or negative attitudes if used in a proper way. Indeed, both [16,45] reveal overall positive values of repayment intervention messages with negative expressions, though their effects may decay with time. [108] indicate that some approach-driven emotional expressions (i.e., offering emotional or physical supports) such as caring and expectations may invoke favorable appraisals of external circumstances, while some avoidance-driven expressions (i.e., not aiming to help people solve the problem) such as anxiety and embarrassment may exert trivial or opposite effects. Yet, the effects may hinge on people's (positive or negative) emotional status. Therefore, our field experiment aims to explore how borrowers with distinct emotions react differently to the positive-expectation and negative-consequence repayment-reminder strategies; especially, we wonder how borrowers with high intensities of positive or negative emotions would perform when they received a negative-consequence reminder message. The answer to this question is unpredictable, as we adopt different emotion measurements (i.e., immediate emotion vs. average emotional tendency), and more importantly, we show that a borrower may present high intensities of happiness and anger simultaneously and that both of these emotion types are associated with default behavior.

In our experimental design, we follow the partnered platform's regular practice of sending repayment reminders one day before borrowers' loan maturity dates. That is, our experimental sample includes only

**Table 4**  
Randomization check.

|                             | Group                      | PosExp Gp<br>N = 40  | NegCons Gp<br>N = 30 | NeutControl Gp<br>N = 40 | F-statistic,<br>p-value |
|-----------------------------|----------------------------|----------------------|----------------------|--------------------------|-------------------------|
| Borrower<br>characteristics | Monthly income level       | 2.667<br>(1.243)     | 2.333<br>(1.184)     | 2.829<br>(1.580)         | $F = 1.17, p = 0.316$   |
|                             | Education level            | 2.429<br>(0.501)     | 2.600<br>(0.498)     | 2.610<br>(0.494)         | $F = 1.67, p = 0.192$   |
|                             | Gender (1 = male)          | 0.643<br>(0.485)     | 0.567<br>(0.504)     | 0.732<br>(0.449)         | $F = 1.06, p = 0.351$   |
|                             | Age                        | 22.357<br>(0.879)    | 22.167<br>(0.874)    | 22.390<br>(1.243)        | $F = 0.46, p = 0.633$   |
| Loan<br>attributes          | Loan term (month)          | 5.833<br>(1.267)     | 5.900<br>(1.029)     | 6.024<br>(1.084)         | $F = 0.30, p = 0.744$   |
|                             | Yearly interest rate (%)   | 14.071<br>(1.332)    | 14.500<br>(1.167)    | 14.171<br>(1.181)        | $F = 1.11, p = 0.335$   |
| Social media<br>behaviors   | Total #blogs posted        | 368.191<br>(560.059) | 352.033<br>(531.633) | 359.634<br>(473.241)     | $F = 0.01, p = 0.992$   |
|                             | Total #followings          | 278.429<br>(264.198) | 280.267<br>(261.268) | 294.098<br>(248.786)     | $F = 0.04, p = 0.957$   |
| Emotional<br>expressions    | Immediate anger            | 0.175<br>(0.192)     | 0.153<br>(0.133)     | 0.159<br>(0.136)         | $F = 0.20, p = 0.820$   |
|                             | Average anger tendency     | 0.172<br>(0.166)     | 0.156<br>(0.102)     | 0.169<br>(0.115)         | $F = 0.14, p = 0.873$   |
|                             | Immediate happiness        | 0.271<br>(0.162)     | 0.230<br>(0.187)     | 0.295<br>(0.194)         | $F = 1.14, p = 0.322$   |
|                             | Average happiness tendency | 0.399<br>(0.169)     | 0.378<br>(0.237)     | 0.406<br>(0.139)         | $F = 0.22, p = 0.805$   |

<sup>a</sup> PosExp, NegCons, and Neut stand for positive-expectation group, negative-consequence group, and neutral (control) group, respectively. Mean values and standard deviations (in parentheses) are reported.

<sup>b</sup> Monthly income level: 1 = USD300 or below, 2 = USD300–USD600, 3 = USD600–USD900, ..., 8 = USD2100–USD2400, 9 = USD2400 or above. Education level: 1 = high school or below, 2 = polytechnics, 3 = undergraduate, 4 = postgraduate.

borrowers who did not fully repay their loans until the day before the loan was due. We randomly assign these borrowers to one of three groups. For the **control group**, we follow the platform's regular practice of sending borrowers *neutral* information via SMS; the message is “Hi XXX, the loan that you received from (platform name) will soon be due. Please login to (platform name) to make your repayment.” Our two treatment messages follow the design in [16]. Specifically, for the **positive-expectation group**, the message is “Hi XXX, the loan that you received from (platform name) will soon be due. Please login to (platform name) to make your repayment. (Platform name) believes that you will do what you promised, so do not let (platform name) down!”. And for the **negative-consequence group**, the message is “Hi XXX, the loan that you received from (platform name) will soon be due. Please login to (platform name) to make your repayment. This is to remind you to repay your loan on time. If you do not, (platform name) will take action to oblige you to repay your loan!”

## 6.2. Experimental data

Our experiment is conducted with borrowers who were granted loans between January and May 2017, soon after our sample period in the secondary data analysis. This greatly mitigates the influence of time across analyses. Consistent with Section 5.3, we focus on borrowers who posted at least once within one month prior to the dates when the reminder messages were sent, and those who became delinquent. We exclude repeated borrowers to avoid unnecessary interference with past experience. Our final experimental sample size for all three groups is 110. We collect and construct identical features to those of the secondary data set, including borrower characteristics, loan attributes, repayment information, borrowers' microblog presence, and emotions. Particularly, we focus on borrowers' emotions at the pre-maturity stage. Table 4 reports the randomization check results along with the value comparisons for the major borrower, loan, social media, and emotion features among the three groups (refer to Table A.3 for complete

statistics for the experimental sample). None of the variables show significant differences across the three groups, as evidenced by the p-values all above 0.05 (ranging from 0.192 to 0.992). This suggests that our experimental randomization is deemed to be successful.

## 6.3. Experimental results

Our experimental analysis is designed to identify heterogeneity among borrowers with diverse emotional statuses. We run our respective analyses at the borrower level (denoted by subscript  $i$ ), and we specify our empirical model in Eq. (3). Similar to Section 5.3, since we focus only on delinquent borrowers, our dependent variable is the default conditional on delinquency, namely,  $\text{Default}_i | \text{Delinquency}_i = 1$ . Besides, taking the control group (to which neutral messages were sent) as the baseline, we use two dummies to respectively indicate the two treatment groups: the positive-expectation group ( $\text{PosExp}_i$ ) and the negative-consequence group ( $\text{NegCons}_i$ ). Since we are interested in how borrowers with different emotions would react to the repayment reminders, we add borrowers' emotions ( $\text{Emotions}_{m_i}$ ) and the interactions between emotions and the treatment indicators (i.e.,  $\text{PosExp}_i \cdot \text{Emotions}_{m_i}$  and  $\text{NegCons}_i \cdot \text{Emotions}_{m_i}$ ).<sup>16</sup> Similarly to Eqs. (1) and (2),  $\text{Emotions}_{m_i}$  are specified as immediate happiness, average happiness tendency, immediate anger, and average anger tendency, respectively, in the estimations. Suffix “m” stood for emotions at the pre-maturity stage. Finally, we control for some borrower-, loan-, and microblog-presence-related factors, which are denoted by  $\vec{X}_i$  and are identical to

<sup>16</sup> We also consider a quadratic form of emotions and their interactions with treatment interventions in the model. We do not find significant parameters for the focal emotion-related variables; we thus do not report these results, though they are available upon request.

**Table 5**  
Repayment reminders and emotions at the pre-maturity stage.

| DV: Default Delinquency = 1   | Immediate emotions  |                    | Average emotional tendencies |                    |
|-------------------------------|---------------------|--------------------|------------------------------|--------------------|
|                               | Anger               | Happiness          | Anger                        | Happiness          |
|                               | (1)                 | (2)                | (3)                          | (4)                |
| PosExp · Emotion_m            | −1.861**<br>(0.839) | −0.779<br>(0.758)  | −2.490**<br>(1.085)          | −1.615*<br>(0.909) |
| NegCons · Emotion_m           | −0.994<br>(1.462)   | 1.730**<br>(0.834) | −0.582<br>(1.796)            | 0.408<br>(0.844)   |
| PosExp                        | 0.366<br>(0.223)    | 0.153<br>(0.265)   | 0.370<br>(0.255)             | 0.557<br>(0.391)   |
| NegCons                       | 0.118<br>(0.242)    | −0.464*<br>(0.269) | 0.046<br>(0.311)             | −0.232<br>(0.393)  |
| Emotion_m                     | 0.657<br>(0.693)    | −0.860*<br>(0.471) | 1.512<br>(0.935)             | 0.062<br>(0.659)   |
| Constant                      | 0.115<br>(2.397)    | −0.794<br>(2.215)  | −0.914<br>(2.430)            | −0.134<br>(2.442)  |
| Controls                      | Yes                 | Yes                | Yes                          | Yes                |
| Loan approval month FE        | Yes                 | Yes                | Yes                          | Yes                |
| Borrowers' living province FE | Yes                 | Yes                | Yes                          | Yes                |
| Mean of Emotion_m             | 0.163               | 0.269              | 0.166                        | 0.396              |
| Adj. R-squared                | 0.213               | 0.289              | 0.189                        | 0.202              |
| N                             | 110                 | 110                | 110                          | 110                |

<sup>a</sup> PosExp and NegCons stand for positive-expectation group and negative-consequence group, respectively; neutral (control) group is used as a baseline.

<sup>b</sup> Models (1) and (3) correspond to using anger as independent variables in Emotion\_m; Models (2) and (4) correspond to happiness. Suffix “m” stands for emotions at the pre-maturity stage.

<sup>c</sup> Standard errors are in parentheses. \*\*\*, \*\* and \* represent statistical significance at the 1%, 5% and 10% levels, respectively.

<sup>d</sup> Control variables include borrower characteristics, loan attributes, and microblog presence. Refer to Table B.3 for full regression results.

those in Eqs. (1) and (2).

$$\begin{aligned}
 \text{Default}_i | (\text{Delinquency}_i = 1) &= \beta_0 + \beta_1 \cdot \text{Emotions\_m}_i + \beta_2 \cdot \text{NegCons}_i \\
 &+ \beta_3 \cdot \text{NegCons}_i \cdot \text{Emotions\_m}_i \\
 &+ \beta_4 \cdot \text{PosExp}_i + \beta_5 \cdot \text{PosExp}_i \cdot \text{Emotions\_m}_i \\
 &+ \bar{X}_i \gamma + \epsilon_i.
 \end{aligned}
 \tag{3}$$

We again apply LPM for the estimation of parameters. Table 5 reports the experimental results. Our experimental analyses yield at least two interesting findings. First, compared with a neutral reminder message, a positive-expectation reminder generally is useful for mitigating default when platform delinquent borrowers are in higher intensities of anger or happiness. Specifically, we find that if delinquent borrowers have been high in anger emotions recently or have been keeping an average anger or happiness tendency prior to the loan maturity dates, a reminder message expressing the platform's or lenders' positive expectation of repayment is effective for promoting borrowers' repayment behavior. For example, a positive-expectation reminder would decrease the default probability by 19.23 percentage points ( $= 0.366 - 1.861 \times 0.30$ ) when a borrower has a strong intensity of anger scored at 0.30. However, we must be cautious, in that for borrowers with low intensity of emotion, the benefits from positive-expectation reminders may decay. Similar patterns are found in the cases of average anger and happiness tendencies. A potential explanation for these results is that having a strong intensity of emotion (in whichever form, happiness or anger) suggests that borrowers are well aware of and form cognitive appraisals about external circumstances, including their financial situations [18], and that therefore, a prompt positive-expectation reminder message is more likely to render them to think about their financial responsibility and persuade them to repay the debts.

Second, we discover that when delinquent borrowers were in a happy mood just prior to the loan maturity date, a negative-consequence reminder message would increase the default probability. For example, a negative-consequence reminder would increase the default probability by 22.80 percentage points ( $= -0.464 + 1.730 \times 0.40$ ) when a borrower has a strong intensity of immediate happiness scored at 0.40. This finding provides evidence for the emotion-related orientation-matching hypothesis (Hypothesis 3), namely, that when borrowers

maintain a high happiness emotion close to the loan maturity date, mismatched communication bringing about deterrence, anxiety, and embarrassment would fail to bring any emotion benefit to them, and thus would lower their self-regulation with respect to financial obligation fulfillment [109].

In Appendix Table B.6, we present results using emotions derived from the large language model (Azure OpenAI GPT-35-turbo-16k model). These results are consistent with those in Table 5, demonstrating the robustness of our findings.

## 7. Discussions and conclusion

Understanding the relationship between individual borrowers' features on a FinTech platform and their credit risk has posed a significant challenge to credit risk management teams in the micro-finance industry, especially in the era of big data. Recent studies have leveraged tremendous sources of information to draw objective individual financial portraits [38]. And yet, borrowers' financial behavior, such as loan default, is more likely to be committed intentionally and be driven by subjective thoughts [5]. Given the absence of empirical insights within the existing literature regarding the connection between subjective characteristics and financial behavior, not to mention leveraging such insights to refine credit risk assessments or formulate more effective loan approval decisions and repayment-reminder intervention strategies, the primary objective of this study is to address this void. While prior research has extensively acknowledged the significance of social media data in augmenting FinTech operations efficiency, and has validated that emotions extracted from online social media can effectively function as an indicator of daily emotions [110], it has paid limited attention to guiding FinTech platform administrative decision-making or gaining a comprehensive understanding of the credit risks associated with borrowers' emotions.

Drawing on an integrated emotional framework that combines appraisal theory of emotions (explaining *why* individuals experience certain emotional reactions to external information) and feelings-as-information theory (clarifying *how* those emotions, once formed, guide decisions under uncertainty), we apply text-mining techniques to automatically extract discrete emotions from borrowers' social media posts. We then use econometric modeling and a field experiment to explore

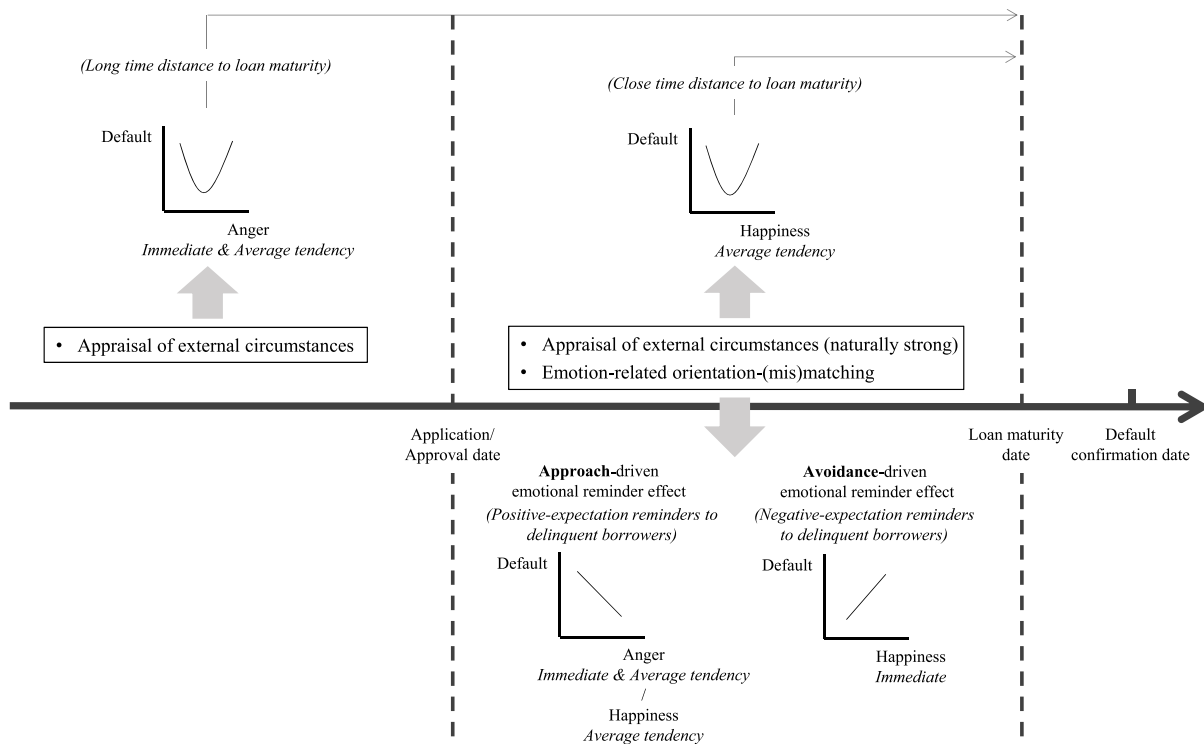


Fig. 3. Visualization of analysis framework and findings.

the impact of these emotions on FinTech platforms' loan-approval processes and repayment intervention strategies at the post-approval and pre-maturity stages. Fig. 3 offers a concise overview of our analytical framework and principal findings. Our study yields a number of non-trivial theoretical and practical implications.

First, as far as we know, this work is among the first to examine the relationships between borrowers' diverse emotions (with a focus on happiness and anger) and their default probability. The literature (e.g., 24) and life experience indicate that positive emotions such as happiness are more likely to be accompanied by moral behavior than negative emotions, as positive emotions originate from favorable appraisals about circumstances [111]. We disclose somewhat counter-intuitive findings within the micro-finance context, namely, that negative emotions such as anger, other than positive emotions, are useful for predicting borrowers' default behavior. More interestingly, we find a U-shaped relationship between anger and default, and determine that such effect holds for both of our two emotion measurements: immediate anger that occurs most recently (within one month) or average anger tendency over a long term (three months) prior to the loan approvals. Our finding suggests that, at the loan-approval decision-making stage when there is a time gap with the fulfillment of the obligation of repayment, borrowers' negative emotions are more informative for predicting their future default.

Second, whereas the literature mostly focuses on FinTech platforms' loan-approval decision-making, this study expands the knowledge of credit risk evaluation by exploring the value of emotions at the loan pre-maturity stage to the prediction of default. We target platform delinquent borrowers who failed to repay on time. Interestingly, we uncover distinct results from those regarding emotions at the pre-approval stage. When the due dates of executing loan repayment are approaching, delinquent borrowers (especially those who became defaulters finally) should in general form passive appraisals (e.g., worry, disappointment) about their financial situations; and negative emotions such as anger and anxiety are aroused consequently. At this time, anger becomes less informative for the prediction of default. Yet, happiness, which opposes a "normal" response to a bad financial situation

(i.e., delinquency), becomes an informative signal. Concretely, a moderate intensity of average happiness tendency indicates a proactive and favorable (e.g., optimistic, sound) appraisal of debts, and delinquent borrowers of moderate average happiness tendency tend to cope with debts positively. In contrast, delinquent borrowers who have been maintaining a strong intensity of happiness over a long term around the pre-maturity period may make unhealthy appraisals (e.g., ignorance, avoidance) about their debts [112], and thus be more likely to default.

Finally, we supplement the emerging repayment intervention strategy and loan recovery literature by discussing a contingent factor concerning borrowers' emotions at the pre-maturity stage. We follow [16] in formulating positive-expectation and negative-consequence repayment reminder messages. Although [16] and other studies (e.g., 44) have documented the effectiveness of these reminder-message framings, our findings suggest that they may work only conditionally on borrowers' different emotional statuses. Specifically, we show that a positive-expectation reminder is useful for decreasing default probability only when borrowers are in high intensities of immediate anger, average anger tendency, or average happiness tendency. A positive-expectation message highlights lenders' emotional or physical supports; we find that as such, it persuades borrowers to repay only when they have formed explicit cognitive appraisals (and thus strong emotions) about their financial situations. An exception to our finding is the case of immediate happiness at the pre-maturity stage. Sending a mismatched negative-consequence reminder message to platform delinquent borrowers with an immediate, or, say, most recent happiness emotion approaching to the loan due dates is likely to cause anxiety, embarrassment, or even a retaliatory intention [45], any of which might result in financial deviance.

In sum, with a special focus on the emotions of happiness and anger, our study explores how subjective features extracted from online social media can be used in comprehensive FinTech platforms' decision-making processes in credit risk management for both the pre-approval and post-approval (especially the pre-maturity) periods. Moreover, While primarily focusing on microloan FinTech platforms, this study's insights may apply to various other microloan services,

including government-to-person, micro-insurance, and mobile money services. Aligning with previous studies [99], specifically, we first show that anger and happiness are not exclusive for a person during the same time period. On a broader level, our rich findings suggest that emotions affect human behavior through a two-step process: first, individuals perform explicit appraisals of their internal and external circumstances based on emotional valence (i.e., positive or negative) and temporal factors (i.e., proximity to responsibility/contract fulfillment); then, these emotions serve as heuristic cues that guide their decision-making and moral behaviors in financial contexts.

Our study's practical implications for FinTech platforms decision makers or lenders are explicit and pronounced. We showcase how practitioners can leverage social media information to run exquisite and effective decision-making support in credit risk management. We reveal what borrowers' emotion types, intensities, and measurements are effective for forecasting their financial default, enhancing decision-making for credit risk evaluation during both the pre-approval and pre-maturity periods. Considering the prevalence of online social media usage, we verify what should be seen as a feasible and low-cost approach to bolstering of credit risk assessment. Moreover, we uncover the moderating effects of repayment reminders and emotions, which can guide FinTech platform managers' implementation of more personalized repayment intervention strategies. Particularly, we suggest that platform managers design positive-expectation and negative-consequence reminder messages with caution; mismatched message framing with borrowers' emotion types or intensities may decay an intervention's effectiveness or even cause it to backfire. Finally, in practical situations, managers in credit assessment and post-loan (post-approval) departments sometimes debate over who should take major responsibility for bad loans. While debates may undermine the operational efficiency of platforms [29], our findings could be crucial for enhancing operational efficiency in both departments through cross-functional coordination. The proposed features, derived from a single source of emotional data, can significantly improve decision-making processes across different stages of the loan lifecycle, thereby enhancing FinTech platform operational efficiency.

The findings of this study should be interpreted in light of its limitations, which nonetheless, offer insights and directions for future research. First, we shed light only on the value of anger and happiness in credit risk management of FinTech platform. Whereas we discover high correlations within the positive or negative categories, exploring other emotion types, whether mentioned or not in this paper, may add more insights into the role of emotions in explaining credit behavior. Second, we take the temporal variance of borrowers' emotions into account by measuring their immediate emotions and average emotion tendencies; however, changes in emotions before and after loan approval might likewise be informative, as they would imply borrowers' updated well-being and socioeconomic status [113]. Future studies with solutions appropriate for endogeneity may enrich our emotion-related variables and verify their value in platform credit risk management. Third, our study focuses exclusively on FinTech platform borrowers who actively use microblogs. As noted by Ge et al. [14], social media usage, engagement, and disclosure behavior may themselves be associated with financial default risk. While we account for data authorization and control for social media presence, including posting frequency, a more thorough examination of the intersection between social media usage patterns and emotional expression could offer deeper insights into borrowers' emotional characteristics and their relevance for credit risk assessment. Fourth, the generalizability of our findings may be constrained by the relatively small sample size of microblog-using borrowers and the potential for strategic content posting on social media, although our rigorous methodology and robust statistical analysis enhance the credibility of our results. Future research could validate and extend our findings by employing larger and more representative samples. Fifth, a possible limitation concerns the gap between felt and expressed emotions. While prior research

indicates that social media posts often mirror users' genuine affective states [72,73], borrowers could still selectively disclose or mask emotions online. However, we have indirectly validated emotional authenticity by averaging multiple posts to mitigate potential disclosure biases. Future work could employ targeted surveys or other data sources to better assess the extent to which borrowers' felt emotions match their digital footprints. Lastly, we use only one source of online social media, and as such, the textual features used to construct emotions may not be generalizable to other social media. Borrowers' social media usage behavior, and writing style, moreover, may vary among countries and cultures. Our study presents valuable opportunities for future studies to explore different contexts of credit risk evaluation, social media sources, and cultural settings.

## CRediT authorship contribution statement

**Zenan Zhou:** Writing – review & editing, Writing – original draft, Methodology, Formal analysis, Resources, Conceptualization. **Zhichen Chen:** Writing – original draft, Visualization, Methodology, Formal analysis, Data curation. **Yingjie Zhang:** Writing – review & editing, Writing – original draft, Resources, Conceptualization. **Tian Lu:** Writing – review & editing, Writing – original draft, Supervision, Resources, Conceptualization. **Xianghua Lu:** Writing – review & editing, Supervision, Resources.

## Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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## Appendix A. Supplementary information on data and variable

### A.1. Correlation analysis results

See [Table A.1](#).

### A.2. Supplementary summary statistics

See [Tables A.2](#) and [A.3](#).

## Appendix B. Supplementary empirical analyses and results

See [Tables B.1–B.6](#).

## Data availability

The authors do not have permission to share data.

Table A.1

Correlation matrices among emotions and default.

| Panel A: Immediate emotions           |          |          |          |          |          |          |          |          |          |          |          |       |       |
|---------------------------------------|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|-------|-------|
|                                       | (1)      | (2)      | (3)      | (4)      | (5)      | (6)      | (7)      | (8)      | (9)      | (10)     | (11)     | (12)  | (13)  |
| (1) Anger_a                           | 1.000    |          |          |          |          |          |          |          |          |          |          |       |       |
| (2) Happiness_a                       | 0.434*** | 1.000    |          |          |          |          |          |          |          |          |          |       |       |
| (3) Surprise_a                        | 0.345*** | 0.837*** | 1.000    |          |          |          |          |          |          |          |          |       |       |
| (4) Disgust_a                         | 0.540*** | 0.475*** | 0.445*** | 1.000    |          |          |          |          |          |          |          |       |       |
| (5) Fear_a                            | 0.245*** | 0.367*** | 0.334*** | 0.365*** | 1.000    |          |          |          |          |          |          |       |       |
| (6) Sadness_a                         | 0.539*** | 0.497*** | 0.458*** | 0.790*** | 0.649*** | 1.000    |          |          |          |          |          |       |       |
| (7) Anger_m                           | 0.084    | 0.136**  | 0.149**  | 0.101*   | 0.075    | 0.328**  | 1.000    |          |          |          |          |       |       |
| (8) Happiness_m                       | 0.109*   | 0.143**  | 0.186*** | 0.120*   | 0.093    | 0.625*** | 0.394*** | 1.000    |          |          |          |       |       |
| (9) Surprise_m                        | 0.094    | 0.177*** | 0.198*** | 0.137**  | 0.079    | 0.590*** | 0.360*** | 0.831*** | 1.000    |          |          |       |       |
| (10) Disgust_m                        | 0.065    | 0.141**  | 0.152**  | 0.134**  | 0.057    | 0.379*** | 0.535*** | 0.417*** | 0.518*** | 1.000    |          |       |       |
| (11) Fear_m                           | 0.118*   | 0.140**  | 0.110*   | 0.094*   | 0.012    | 0.187*** | 0.294*** | 0.233*** | 0.322*** | 0.312*** | 1.000    |       |       |
| (12) Sadness_m                        | 0.076    | 0.144**  | 0.126**  | 0.131**  | 0.036    | 0.343*** | 0.545*** | 0.427*** | 0.527*** | 0.765*** | 0.553*** | 1.000 |       |
| (13) Default                          | -0.100*  | -0.119*  | -0.097*  | -0.071   | -0.031   | -0.017   | -0.038   | 0.036    | 0.064    | 0.035    | 0.022    | 0.014 | 1.000 |
| Panel B: Average emotional tendencies |          |          |          |          |          |          |          |          |          |          |          |       |       |
|                                       | (1)      | (2)      | (3)      | (4)      | (5)      | (6)      | (7)      | (8)      | (9)      | (10)     | (11)     | (12)  | (13)  |
| (1) Anger_a                           | 1.000    |          |          |          |          |          |          |          |          |          |          |       |       |
| (2) Happiness_a                       | 0.462*** | 1.000    |          |          |          |          |          |          |          |          |          |       |       |
| (3) Surprise_a                        | 0.411*** | 0.867*** | 1.000    |          |          |          |          |          |          |          |          |       |       |
| (4) Disgust_a                         | 0.601*** | 0.460*** | 0.419*** | 1.000    |          |          |          |          |          |          |          |       |       |
| (5) Fear_a                            | 0.291*** | 0.363*** | 0.315*** | 0.224*** | 1.000    |          |          |          |          |          |          |       |       |
| (6) Sadness_a                         | 0.638*** | 0.500*** | 0.461*** | 0.808*** | 0.504*** | 1.000    |          |          |          |          |          |       |       |
| (7) Anger_m                           | 0.128**  | 0.134**  | 0.150**  | 0.118*   | 0.086    | 0.353*** | 1.000    |          |          |          |          |       |       |
| (8) Happiness_m                       | 0.148**  | 0.218*** | 0.253*** | 0.133**  | 0.203*** | 0.679*** | 0.416*** | 1.000    |          |          |          |       |       |
| (9) Surprise_m                        | 0.172*** | 0.219*** | 0.272*** | 0.171*** | 0.184*** | 0.632*** | 0.367*** | 0.888**  | 1.000    |          |          |       |       |
| (10) Disgust_m                        | 0.078    | 0.098*   | 0.156**  | 0.166*** | 0.121*   | 0.506*** | 0.592*** | 0.514*** | 0.520*** | 1.000    |          |       |       |
| (11) Fear_m                           | 0.186*** | 0.103*   | 0.139**  | 0.095*   | 0.066    | 0.309*** | 0.279*** | 0.377*** | 0.430*** | 0.304*** | 1.000    |       |       |
| (12) Sadness_m                        | 0.099*   | 0.092    | 0.133**  | 0.136**  | 0.148**  | 0.396*** | 0.578*** | 0.512*** | 0.557*** | 0.787*** | 0.600*** | 1.000 |       |
| (13) Default                          | -0.062   | -0.101*  | -0.074   | -0.036   | -0.021   | 0.028    | -0.022   | 0.058    | 0.059    | 0.083    | -0.004   | 0.057 | 1.000 |

<sup>a</sup> Since we consider and show emotions at both pre-approval and pre-maturity stages in this table, we report correlation matrices on the subsample of 437 delinquent (including defaulted) borrowers; refer to Section 5.3 for details.

<sup>b</sup> Suffix “a” stands for emotions at the pre-approval stage, and suffix “m” stands for emotions at the pre-maturity stage.

<sup>c</sup> Values in shaded cells are correlations of the same emotion type’s presence between the pre-approval and pre-maturity stages.

<sup>d</sup> \*\*\*, \*\* and \* represent statistical significance at the 1%, 5% and 10% levels, respectively.

Table A.2

Descriptive statistics for delinquent borrowers.

| N=437                                             | (1)     | (2)      | (3)  | (4) |
|---------------------------------------------------|---------|----------|------|-----|
|                                                   | Mean    | Std. Dev | Max  | Min |
| <i>Repayment performance</i>                      |         |          |      |     |
| Default (conditional on delinquency)              | 0.382   | 0.487    | 1    | 0   |
| <i>Emotions at the pre-maturity stage</i>         |         |          |      |     |
| Immediate anger                                   | 0.160   | 0.145    | 1    | 0   |
| Average anger tendency                            | 0.199   | 0.134    | 1    | 0   |
| Immediate happiness                               | 0.180   | 0.125    | 1    | 0   |
| Average happiness tendency                        | 0.281   | 0.143    | 1    | 0   |
| <i>Borrower characteristics</i>                   |         |          |      |     |
| Monthly income level                              | 2.223   | 1.252    | 6    | 1   |
| Education level                                   | 2.601   | 0.490    | 3    | 2   |
| Gender (1 = male)                                 | 0.772   | 0.420    | 1    | 0   |
| Age                                               | 23.107  | 1.062    | 30   | 21  |
| #Children                                         | 0.002   | 0.048    | 1    | 0   |
| Marital status (1 = in a marriage)                | 0.005   | 0.067    | 1    | 0   |
| House ownership (1 = self-own)                    | 0.118   | 0.324    | 1    | 0   |
| <i>Loan attributes</i>                            |         |          |      |     |
| Loan amount (USD)                                 | 455.961 | 41.704   | 465  | 155 |
| Loan purpose (1 = used for emergency)             | 0.387   | 0.488    | 1    | 0   |
| Loan term (month)                                 | 6.451   | 2.573    | 21   | 2   |
| Loan history (1 = borrowed on the company before) | 0.162   | 0.369    | 1    | 0   |
| Yearly interest rate (%)                          | 14.091  | 0.887    | 16   | 12  |
| <i>Microblog presence (static)</i>                |         |          |      |     |
| Total #blogs posted                               | 456.800 | 690.007  | 2817 | 26  |
| Total #fans                                       | 559.713 | 1178.028 | 4951 | 17  |
| Total #followings                                 | 269.770 | 201.057  | 796  | 37  |
| Total #reposts received                           | 59.376  | 165.996  | 709  | 0   |
| Total #comments received                          | 222.535 | 438.173  | 1807 | 0   |

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**Table A.2** (continued).

| N=437                          | (1)     | (2)      | (3)  | (4) |
|--------------------------------|---------|----------|------|-----|
|                                | Mean    | Std. Dev | Max  | Min |
| Total #“likes” received        | 257.205 | 498.761  | 2124 | 1   |
| Max. #reposts a blog received  | 3.305   | 4.570    | 19   | 0   |
| Max. #comments a blog received | 5.576   | 3.141    | 9    | 0   |
| Max. #“likes” a blog received  | 5.141   | 2.916    | 9    | 1   |

<sup>a</sup> This table is based on the subsample of delinquent borrowers, namely borrowers who have fully repaid their loans in advance and on time are excluded.

<sup>b</sup> Monthly income level: 1 = USD300 or below, 2 = USD300–USD600, 3 = USD600–USD900, ..., 8 = USD2100–USD2400, 9 = USD2400 or above. Education level: 1 = high school or below, 2 = polytechnics, 3 = undergraduate, 4 = postgraduate.

**Table A.3**

Descriptive statistics for experimental sample.

| N=110                                             | (1)     | (2)      | (3)  | (4) |
|---------------------------------------------------|---------|----------|------|-----|
|                                                   | Mean    | Std. Dev | Max  | Min |
| <i>Emotions at the pre-maturity stage</i>         |         |          |      |     |
| Immediate anger                                   | 0.163   | 0.157    | 1    | 0   |
| Average anger tendency                            | 0.166   | 0.133    | 1    | 0   |
| Immediate happiness                               | 0.269   | 0.181    | 1    | 0   |
| Average happiness tendency                        | 0.396   | 0.179    | 1    | 0   |
| <i>Borrowers' Demographics</i>                    |         |          |      |     |
| Monthly income level                              | 2.637   | 1.363    | 7    | 1   |
| Education level                                   | 2.540   | 0.501    | 3    | 2   |
| Gender (1 = male)                                 | 0.655   | 0.478    | 1    | 0   |
| Age                                               | 22.319  | 1.020    | 26   | 21  |
| #Children                                         | 0.009   | 0.094    | 1    | 0   |
| Marital status (1 = in a marriage)                | 0.009   | 0.094    | 1    | 0   |
| House ownership (1 = self-own)                    | 0.274   | 0.448    | 1    | 0   |
| <i>Loan characteristics</i>                       |         |          |      |     |
| Loan amount (USD)                                 | 462.531 | 18.591   | 465  | 310 |
| Loan purpose (1 = used for emergency)             | 0.381   | 0.488    | 1    | 0   |
| Loan term (month)                                 | 5.920   | 1.135    | 7    | 2   |
| Loan history (1 = borrowed on the company before) | 0.177   | 0.383    | 1    | 0   |
| Yearly interest rate (%)                          | 14.221  | 1.237    | 16   | 12  |
| <i>Microblog presence (static)</i>                |         |          |      |     |
| Total #blogs posted                               | 360.442 | 517.913  | 2064 | 15  |
| Total #fans                                       | 298.212 | 542.374  | 2335 | 20  |
| Total #followings                                 | 283.168 | 255.182  | 932  | 35  |
| Total #reposts received                           | 25.982  | 61.159   | 245  | 0   |
| Total #comments received                          | 106.354 | 212.157  | 857  | 1   |
| Total #“likes” received                           | 111.681 | 162.716  | 584  | 1   |
| Max. #reposts a blog received                     | 2.690   | 3.648    | 14   | 0   |
| Max. #comments a blog received                    | 4.752   | 3.181    | 9    | 1   |
| Max. #“likes” a blog received                     | 4.248   | 2.600    | 9    | 1   |

<sup>a</sup> Monthly income level: 1 = USD300 or below, 2 = USD300–USD600, 3 = USD600–USD900, ..., 8 = USD2100–USD2400, 9 = USD2400 or above. Education level: 1 = high school or below, 2 = polytechnics, 3 = undergraduate, 4 = postgraduate.

**Table B.1**

Full regression table: Emotions at the pre-approval stage and loan default.

| DV: Default                      | Immediate emotions  |                      |                     |                     | Average emotional tendencies |                      |                     |                     |
|----------------------------------|---------------------|----------------------|---------------------|---------------------|------------------------------|----------------------|---------------------|---------------------|
|                                  | Anger               |                      | Happiness           |                     | Anger                        |                      | Happiness           |                     |
|                                  | (1)                 | (2)                  | (3)                 | (4)                 | (5)                          | (6)                  | (7)                 | (8)                 |
| Emotion_a                        | −0.047<br>(0.063)   | −0.392***<br>(0.136) | −0.051<br>(0.060)   | 0.202<br>(0.157)    | −0.045<br>(0.076)            | −0.506***<br>(0.158) | −0.025<br>(0.072)   | 0.040<br>(0.192)    |
| Emotion_a_square                 |                     | 0.593***<br>(0.206)  |                     | −0.391<br>(0.244)   |                              | 0.815***<br>(0.245)  |                     | −0.103<br>(0.282)   |
| Education level                  | 0.125***<br>(0.027) | 0.125***<br>(0.027)  | 0.126***<br>(0.027) | 0.124***<br>(0.027) | 0.126***<br>(0.027)          | 0.126***<br>(0.027)  | 0.126***<br>(0.027) | 0.126***<br>(0.027) |
| Education level · Income level 2 | −0.027*<br>(0.017)  | −0.028*<br>(0.017)   | −0.027*<br>(0.017)  | −0.026*<br>(0.015)  | −0.028*<br>(0.017)           | −0.028*<br>(0.016)   | −0.028*<br>(0.017)  | −0.028*<br>(0.017)  |
| Education level · Income level 3 | −0.020              | −0.023               | −0.020              | −0.018              | −0.021                       | −0.025               | −0.021              | −0.021              |

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Table B.1 (continued).

| DV: Default                      | Immediate emotions |           |           |           | Average emotional tendencies |           |           |           |
|----------------------------------|--------------------|-----------|-----------|-----------|------------------------------|-----------|-----------|-----------|
|                                  | Anger              |           | Happiness |           | Anger                        |           | Happiness |           |
|                                  | (1)                | (2)       | (3)       | (4)       | (5)                          | (6)       | (7)       | (8)       |
|                                  | (0.027)            | (0.027)   | (0.027)   | (0.027)   | (0.027)                      | (0.027)   | (0.027)   | (0.027)   |
| Education level · Income level 4 | −0.053             | −0.057    | −0.053    | −0.050    | −0.054                       | −0.060    | −0.054    | −0.054    |
|                                  | (0.040)            | (0.040)   | (0.040)   | (0.040)   | (0.040)                      | (0.040)   | (0.040)   | (0.040)   |
| Education level · Income level 5 | −0.133**           | −0.137**  | −0.133**  | −0.127**  | −0.135**                     | −0.141**  | −0.135**  | −0.135**  |
|                                  | (0.056)            | (0.056)   | (0.056)   | (0.056)   | (0.056)                      | (0.056)   | (0.056)   | (0.056)   |
| Education level · Income level 6 | −0.139**           | −0.148**  | −0.139**  | −0.133*   | −0.141**                     | −0.150**  | −0.141**  | −0.140**  |
|                                  | (0.068)            | (0.068)   | (0.068)   | (0.068)   | (0.068)                      | (0.068)   | (0.068)   | (0.068)   |
| Education level · Income level 7 | −0.131*            | −0.137*   | −0.130*   | −0.125*   | −0.133*                      | −0.141*   | −0.133*   | −0.132*   |
|                                  | (0.074)            | (0.077)   | (0.074)   | (0.070)   | (0.075)                      | (0.079)   | (0.075)   | (0.075)   |
| Education level · Income level 8 | −0.192**           | −0.199*   | −0.192**  | −0.185**  | −0.193**                     | −0.206*   | −0.194**  | −0.193**  |
|                                  | (0.092)            | (0.121)   | (0.091)   | (0.089)   | (0.092)                      | (0.121)   | (0.092)   | (0.092)   |
| Education level · Income level 9 | −0.191**           | −0.209**  | −0.189**  | −0.183**  | −0.188**                     | −0.191**  | −0.189**  | −0.190**  |
|                                  | (0.091)            | (0.100)   | (0.089)   | (0.087)   | (0.089)                      | (0.091)   | (0.089)   | (0.089)   |
| Gender                           | 0.059***           | 0.060***  | 0.059***  | 0.060***  | 0.059***                     | 0.056**   | 0.059***  | 0.059***  |
|                                  | (0.022)            | (0.022)   | (0.022)   | (0.022)   | (0.022)                      | (0.022)   | (0.022)   | (0.022)   |
| Age                              | 0.005              | 0.003     | 0.005     | 0.005     | 0.005                        | 0.004     | 0.005     | 0.005     |
|                                  | (0.009)            | (0.009)   | (0.009)   | (0.009)   | (0.009)                      | (0.009)   | (0.009)   | (0.009)   |
| #Children                        | −0.047             | −0.042    | −0.049    | −0.050    | −0.046                       | −0.040    | −0.047    | −0.047    |
|                                  | (0.077)            | (0.077)   | (0.077)   | (0.077)   | (0.077)                      | (0.077)   | (0.077)   | (0.077)   |
| Marital status                   | 0.192**            | 0.183**   | 0.191**   | 0.188**   | 0.191**                      | 0.178*    | 0.190**   | 0.190**   |
|                                  | (0.092)            | (0.092)   | (0.092)   | (0.092)   | (0.092)                      | (0.092)   | (0.092)   | (0.092)   |
| House ownership                  | 0.057              | 0.058     | 0.057     | 0.058     | 0.056                        | 0.059     | 0.057     | 0.057     |
|                                  | (0.047)            | (0.047)   | (0.047)   | (0.047)   | (0.047)                      | (0.047)   | (0.047)   | (0.047)   |
| Loan amount                      | 0.000***           | 0.000***  | 0.000***  | 0.000***  | 0.000***                     | 0.000***  | 0.000***  | 0.000***  |
|                                  | (0.000)            | (0.000)   | (0.000)   | (0.000)   | (0.000)                      | (0.000)   | (0.000)   | (0.000)   |
| Loan purpose                     | 0.016              | 0.017     | 0.016     | 0.017     | 0.016                        | 0.018     | 0.016     | 0.016     |
|                                  | (0.020)            | (0.020)   | (0.020)   | (0.020)   | (0.020)                      | (0.020)   | (0.020)   | (0.020)   |
| Loan term                        | 0.011***           | 0.011***  | 0.011***  | 0.011***  | 0.011***                     | 0.011***  | 0.011***  | 0.011***  |
|                                  | (0.004)            | (0.004)   | (0.004)   | (0.004)   | (0.004)                      | (0.004)   | (0.004)   | (0.004)   |
| Loan history                     | −0.049*            | −0.048*   | −0.050*   | −0.051*   | −0.049*                      | −0.050*   | −0.050*   | −0.050*   |
|                                  | (0.027)            | (0.027)   | (0.027)   | (0.027)   | (0.027)                      | (0.027)   | (0.027)   | (0.027)   |
| Yearly interest rate             | 0.022**            | 0.023**   | 0.022**   | 0.023**   | 0.022**                      | 0.022**   | 0.022**   | 0.023**   |
|                                  | (0.010)            | (0.010)   | (0.010)   | (0.010)   | (0.010)                      | (0.010)   | (0.010)   | (0.010)   |
| Total #blogs posted              | −0.000*            | −0.000    | −0.000*   | −0.000*   | −0.000*                      | −0.000    | −0.000*   | −0.000*   |
|                                  | (0.000)            | (0.000)   | (0.000)   | (0.000)   | (0.000)                      | (0.000)   | (0.000)   | (0.000)   |
| Total #fans                      | 0.000*             | 0.000*    | 0.000*    | 0.000*    | 0.000*                       | 0.000     | 0.000*    | 0.000*    |
|                                  | (0.000)            | (0.000)   | (0.000)   | (0.000)   | (0.000)                      | (0.000)   | (0.000)   | (0.000)   |
| Total #followings                | 0.000              | 0.000     | 0.000     | 0.000     | 0.000                        | 0.000     | 0.000     | 0.000     |
|                                  | (0.000)            | (0.000)   | (0.000)   | (0.000)   | (0.000)                      | (0.000)   | (0.000)   | (0.000)   |
| Total #reposts received          | −0.001             | −0.001    | −0.001    | −0.001    | −0.001                       | −0.001    | −0.001    | −0.001    |
|                                  | (0.001)            | (0.001)   | (0.001)   | (0.001)   | (0.001)                      | (0.001)   | (0.001)   | (0.001)   |
| Total #comments received         | 0.000              | 0.000     | 0.000     | 0.000     | 0.000                        | 0.000     | 0.000     | 0.000     |
|                                  | (0.000)            | (0.000)   | (0.000)   | (0.000)   | (0.000)                      | (0.000)   | (0.000)   | (0.000)   |
| Total #“likes” received          | −0.000             | −0.000    | −0.000    | −0.000    | −0.000                       | −0.000    | −0.000    | −0.000    |
|                                  | (0.000)            | (0.000)   | (0.000)   | (0.000)   | (0.000)                      | (0.000)   | (0.000)   | (0.000)   |
| Max. #reposts a blog received    | 0.011*             | 0.011*    | 0.011*    | 0.011*    | 0.012**                      | 0.013**   | 0.012**   | 0.012**   |
|                                  | (0.006)            | (0.006)   | (0.006)   | (0.006)   | (0.006)                      | (0.006)   | (0.006)   | (0.006)   |
| Max. #comments a blog received   | −0.003             | −0.003    | −0.003    | −0.003    | −0.003                       | −0.002    | −0.003    | −0.003    |
|                                  | (0.004)            | (0.004)   | (0.004)   | (0.004)   | (0.004)                      | (0.004)   | (0.004)   | (0.004)   |
| Max. #“likes” a blog received    | 0.002              | 0.002     | 0.002     | 0.002     | 0.002                        | 0.002     | 0.002     | 0.002     |
|                                  | (0.005)            | (0.005)   | (0.005)   | (0.005)   | (0.005)                      | (0.005)   | (0.005)   | (0.005)   |
| Constant                         | −0.983***          | −0.923*** | −0.980*** | −1.018*** | −0.987***                    | −0.920*** | −0.990*** | −0.998*** |
|                                  | (0.303)            | (0.303)   | (0.303)   | (0.304)   | (0.303)                      | (0.303)   | (0.303)   | (0.304)   |
| Loan application month FE        | Yes                | Yes       | Yes       | Yes       | Yes                          | Yes       | Yes       | Yes       |
| Living province FE               | Yes                | Yes       | Yes       | Yes       | Yes                          | Yes       | Yes       | Yes       |
| Mean of Emotion_a                | 0.150              | 0.150     | 0.238     | 0.238     | 0.152                        | 0.152     | 0.243     | 0.243     |
| Adj. R-squared                   | 0.049              | 0.053     | 0.049     | 0.050     | 0.049                        | 0.055     | 0.049     | 0.049     |
| N                                | 1943               | 1943      | 1943      | 1943      | 1943                         | 1943      | 1943      | 1943      |

<sup>a</sup> Models (1), (2), (5), and (6) correspond to models using anger as independent variables in Emotion\_a and Emotion\_a\_square; Models (3), (4), (7), and (8) use happiness. Suffix “a” stands for emotions at the pre-approval stage. For Income level, the category Income level = 1 serves as the baseline.

<sup>b</sup> Standard errors are in parentheses. \*\*\*, \*\* and \* represent statistical significance at the 1%, 5% and 10% levels, respectively.

**Table B.2**

Full regression table: Emotions at the pre-maturity stage and loan default.

| DV: Default Delinquency = 1      | Immediate emotions  |                     |                     |                     | Average emotional tendencies |                     |                     |                     |
|----------------------------------|---------------------|---------------------|---------------------|---------------------|------------------------------|---------------------|---------------------|---------------------|
|                                  | Anger               |                     | Happiness           |                     | Anger                        |                     | Happiness           |                     |
|                                  | (1)                 | (2)                 | (3)                 | (4)                 | (5)                          | (6)                 | (7)                 | (8)                 |
| Emotion_m                        | −0.099<br>(0.165)   | −0.181<br>(0.365)   | 0.139<br>(0.195)    | 0.166<br>(0.430)    | −0.142<br>(0.185)            | −0.691<br>(0.427)   | 0.195<br>(0.172)    | −0.856*<br>(0.459)  |
| Emotion_m_square                 |                     | 0.147<br>(0.580)    |                     | −0.048<br>(0.674)   |                              | 0.919<br>(0.625)    |                     | 1.487**<br>(0.602)  |
| Education level                  | 0.197***<br>(0.073) | 0.198***<br>(0.073) | 0.197***<br>(0.073) | 0.197***<br>(0.073) | 0.193***<br>(0.074)          | 0.190**<br>(0.074)  | 0.190**<br>(0.073)  | 0.189***<br>(0.073) |
| Education level · Income level 2 | −0.015<br>(0.047)   | −0.015<br>(0.047)   | −0.014<br>(0.047)   | −0.014<br>(0.047)   | −0.011<br>(0.048)            | −0.009<br>(0.048)   | −0.008<br>(0.047)   | −0.005<br>(0.047)   |
| Education level · Income level 3 | −0.021<br>(0.084)   | −0.022<br>(0.084)   | −0.024<br>(0.084)   | −0.025<br>(0.084)   | −0.020<br>(0.085)            | −0.015<br>(0.085)   | −0.022<br>(0.085)   | −0.019<br>(0.084)   |
| Education level · Income level 4 | −0.046<br>(0.122)   | −0.047<br>(0.123)   | −0.050<br>(0.123)   | −0.050<br>(0.123)   | −0.045<br>(0.125)            | −0.038<br>(0.125)   | −0.046<br>(0.124)   | −0.041<br>(0.123)   |
| Education level · Income level 5 | −0.192**<br>(0.090) | −0.195**<br>(0.092) | −0.198**<br>(0.093) | −0.198**<br>(0.093) | −0.200**<br>(0.095)          | −0.194**<br>(0.091) | −0.194**<br>(0.091) | −0.192**<br>(0.090) |
| Education level · Income level 6 | −0.192**<br>(0.089) | −0.194**<br>(0.090) | −0.192**<br>(0.089) | −0.193**<br>(0.090) | −0.194**<br>(0.090)          | −0.191**<br>(0.089) | −0.187**<br>(0.087) | −0.189**<br>(0.088) |
| Gender                           | −0.006<br>(0.056)   | −0.008<br>(0.057)   | −0.005<br>(0.056)   | −0.005<br>(0.056)   | −0.002<br>(0.056)            | −0.009<br>(0.056)   | −0.002<br>(0.056)   | −0.009<br>(0.056)   |
| Age                              | 0.001<br>(0.026)    | 0.001<br>(0.026)    | −0.002<br>(0.026)   | −0.002<br>(0.026)   | −0.003<br>(0.026)            | −0.003<br>(0.026)   | −0.006<br>(0.026)   | −0.004<br>(0.026)   |
| #Children                        | −0.512<br>(0.756)   | −0.504<br>(0.758)   | −0.610<br>(0.756)   | −0.613<br>(0.758)   | −0.675<br>(0.751)            | −0.677<br>(0.750)   | −0.805<br>(0.754)   | −0.802<br>(0.749)   |
| Marital status                   | 0.688<br>(0.498)    | 0.690<br>(0.499)    | 0.722<br>(0.498)    | 0.723<br>(0.499)    | 0.689<br>(0.500)             | 0.712<br>(0.500)    | 0.744<br>(0.500)    | 0.763<br>(0.497)    |
| House ownership                  | −0.025<br>(0.078)   | −0.024<br>(0.078)   | −0.024<br>(0.078)   | −0.024<br>(0.079)   | −0.024<br>(0.079)            | −0.024<br>(0.078)   | −0.023<br>(0.078)   | −0.003<br>(0.078)   |
| Loan amount                      | 0.000<br>(0.000)    | 0.000<br>(0.000)    | 0.000<br>(0.000)    | 0.000<br>(0.000)    | 0.000<br>(0.000)             | 0.000<br>(0.000)    | 0.000<br>(0.000)    | 0.000<br>(0.000)    |
| Loan purpose                     | 0.044<br>(0.048)    | 0.044<br>(0.048)    | 0.043<br>(0.048)    | 0.043<br>(0.048)    | 0.040<br>(0.048)             | 0.040<br>(0.048)    | 0.044<br>(0.048)    | 0.045<br>(0.047)    |
| Loan term                        | 0.013<br>(0.009)    | 0.013<br>(0.009)    | 0.013<br>(0.009)    | 0.013<br>(0.009)    | 0.015<br>(0.009)             | 0.013<br>(0.009)    | 0.014<br>(0.009)    | 0.012<br>(0.009)    |
| Loan history                     | −0.064<br>(0.067)   | −0.066<br>(0.067)   | −0.055<br>(0.067)   | −0.054<br>(0.067)   | −0.067<br>(0.067)            | −0.069<br>(0.067)   | −0.055<br>(0.067)   | −0.066<br>(0.067)   |
| Yearly interest rate             | 0.039<br>(0.027)    | 0.039<br>(0.027)    | 0.041<br>(0.027)    | 0.041<br>(0.027)    | 0.041<br>(0.027)             | 0.041<br>(0.027)    | 0.044*<br>(0.027)   | 0.039<br>(0.027)    |
| Total #blogs posted              | −0.000<br>(0.000)   | −0.000<br>(0.000)   | −0.000<br>(0.000)   | −0.000<br>(0.000)   | −0.000<br>(0.000)            | −0.000<br>(0.000)   | −0.000<br>(0.000)   | −0.000<br>(0.000)   |
| Total #fans                      | 0.000<br>(0.000)    | 0.000<br>(0.000)    | 0.000<br>(0.000)    | 0.000<br>(0.000)    | 0.000<br>(0.000)             | 0.000<br>(0.000)    | 0.000<br>(0.000)    | 0.000<br>(0.000)    |
| Total #followings                | 0.000<br>(0.000)    | 0.000<br>(0.000)    | 0.000<br>(0.000)    | 0.000<br>(0.000)    | 0.000*<br>(0.000)            | 0.000*<br>(0.000)   | 0.000*<br>(0.000)   | 0.000*<br>(0.000)   |
| Total #reposts received          | 0.000<br>(0.001)    | 0.000<br>(0.001)    | 0.000<br>(0.001)    | 0.000<br>(0.001)    | −0.000<br>(0.001)            | −0.000<br>(0.001)   | −0.001<br>(0.001)   | −0.001<br>(0.001)   |
| Total #comments received         | −0.000**<br>(0.000) | −0.000**<br>(0.000) | −0.000**<br>(0.000) | −0.000**<br>(0.000) | −0.000*<br>(0.000)           | −0.000*<br>(0.000)  | −0.000*<br>(0.000)  | −0.000*<br>(0.000)  |
| Total #“likes”                   | 0.000<br>(0.000)    | 0.000<br>(0.000)    | 0.000<br>(0.000)    | 0.000<br>(0.000)    | 0.000<br>(0.000)             | 0.000<br>(0.000)    | 0.000<br>(0.000)    | 0.000<br>(0.000)    |
| Max. #reposts a blog received    | 0.013<br>(0.011)    | 0.013<br>(0.011)    | 0.013<br>(0.011)    | 0.013<br>(0.011)    | 0.018<br>(0.013)             | 0.019<br>(0.013)    | 0.019<br>(0.013)    | 0.020<br>(0.012)    |
| Max. #comments a blog received   | 0.013<br>(0.011)    | 0.013<br>(0.012)    | 0.013<br>(0.011)    | 0.013<br>(0.012)    | 0.014<br>(0.012)             | 0.014<br>(0.012)    | 0.014<br>(0.012)    | 0.013<br>(0.012)    |
| Max. #“likes” a blog received    | −0.006<br>(0.013)   | −0.006<br>(0.013)   | −0.005<br>(0.013)   | −0.004<br>(0.013)   | −0.005<br>(0.013)            | −0.007<br>(0.013)   | −0.003<br>(0.013)   | −0.004<br>(0.013)   |
| Constant                         | −1.097<br>(0.875)   | −1.082<br>(0.878)   | −1.109<br>(0.875)   | −1.118<br>(0.885)   | −1.015<br>(0.877)            | −0.918<br>(0.878)   | −1.085<br>(0.877)   | −0.874<br>(0.876)   |
| Loan application month FE        | Yes                 | Yes                 | Yes                 | Yes                 | Yes                          | Yes                 | Yes                 | Yes                 |
| Living province FE               | Yes                 | Yes                 | Yes                 | Yes                 | Yes                          | Yes                 | Yes                 | Yes                 |
| Mean of Emotion_m                | 0.160               | 0.160               | 0.180               | 0.180               | 0.199                        | 0.199               | 0.281               | 0.281               |
| Adj. R-squared                   | 0.118               | 0.116               | 0.119               | 0.116               | 0.113                        | 0.116               | 0.115               | 0.127               |
| N                                | 437                 | 437                 | 437                 | 437                 | 437                          | 437                 | 437                 | 437                 |

<sup>a</sup> Models (1), (2), (5), and (6) correspond to using anger as independent variables in Emotion\_m and Emotion\_m\_square; Models (3), (4), (7), and (8) correspond to happiness. Suffix “m” stands for emotions at the pre-maturity stage. For Income level, the category Income level = 1 serves as the baseline.

<sup>b</sup> Standard errors are in parentheses. \*\*\*, \*\* and \* represent statistical significance at the 1%, 5% and 10% levels, respectively.

<sup>c</sup> Browsers with an income level of 7, 8, or 9 are not delinquent in our sample.

**Table B.3**

Full regression table: Repayment reminders and emotions at the pre-maturity stage.

| DV: Default Delinquency = 1      | Immediate emotions  |                     | Average emotional tendencies |                     |
|----------------------------------|---------------------|---------------------|------------------------------|---------------------|
|                                  | Anger               | Happiness           | Anger                        | Happiness           |
|                                  | (1)                 | (2)                 | (3)                          | (4)                 |
| PosExp · Emotion_m               | −1.861**<br>(0.839) | −0.779<br>(0.758)   | −2.490**<br>(1.085)          | −1.615*<br>(0.909)  |
| NegCons · Emotion_m              | −0.994<br>(1.462)   | 1.730**<br>(0.834)  | −0.582<br>(1.796)            | 0.408<br>(0.844)    |
| PosExp                           | 0.366<br>(0.223)    | 0.153<br>(0.265)    | 0.370<br>(0.255)             | 0.557<br>(0.391)    |
| NegCons                          | 0.118<br>(0.242)    | −0.464*<br>(0.269)  | 0.046<br>(0.311)             | −0.232<br>(0.393)   |
| Emotion_m                        | 0.657<br>(0.693)    | −0.860*<br>(0.471)  | 1.512<br>(0.935)             | 0.062<br>(0.659)    |
| Education level                  | 0.294<br>(0.197)    | 0.189<br>(0.190)    | 0.221<br>(0.202)             | 0.135<br>(0.209)    |
| Education level · Income level 2 | −0.141<br>(0.125)   | −0.091<br>(0.116)   | −0.064<br>(0.130)            | −0.056<br>(0.126)   |
| Education level · Income level 3 | −0.236<br>(0.222)   | −0.119<br>(0.210)   | −0.078<br>(0.225)            | −0.024<br>(0.228)   |
| Education level · Income level 4 | −0.423*<br>(0.247)  | −0.354*<br>(0.205)  | −0.132*<br>(0.071)           | −0.147*<br>(0.079)  |
| Education level · Income level 5 | −0.646**<br>(0.321) | −0.430**<br>(0.211) | −0.309**<br>(0.148)          | −0.185**<br>(0.086) |
| Education level · Income level 6 | −0.555**<br>(0.272) | −0.422**<br>(0.204) | −0.115*<br>(0.061)           | −0.065*<br>(0.032)  |
| Education level · Income level 7 | −0.637**<br>(0.314) | −0.516*<br>(0.302)  | −0.195*<br>(0.109)           | −0.155**<br>(0.070) |
| Gender                           | 0.264**<br>(0.123)  | 0.320***<br>(0.116) | 0.243*<br>(0.131)            | 0.247*<br>(0.124)   |
| Age                              | −0.123*<br>(0.069)  | −0.058<br>(0.063)   | −0.083<br>(0.070)            | −0.059<br>(0.065)   |
| #Children                        | −0.304<br>(0.651)   | −0.301<br>(0.637)   | −0.169<br>(0.682)            | −0.094<br>(0.684)   |
| House ownership                  | 0.180<br>(0.133)    | 0.186<br>(0.125)    | 0.106<br>(0.139)             | 0.088<br>(0.134)    |
| Loan amount                      | 0.000<br>(0.001)    | 0.000<br>(0.001)    | 0.000<br>(0.001)             | 0.000<br>(0.001)    |
| Loan purpose                     | 0.010<br>(0.129)    | −0.012<br>(0.119)   | 0.000<br>(0.127)             | −0.029<br>(0.125)   |
| Loan term                        | 0.135**<br>(0.059)  | 0.096*<br>(0.056)   | 0.159**<br>(0.060)           | 0.118*<br>(0.060)   |
| Loan history                     | −0.035<br>(0.170)   | −0.075<br>(0.158)   | −0.076<br>(0.172)            | −0.100<br>(0.159)   |
| Yearly interest rate             | 0.075<br>(0.045)    | 0.025<br>(0.042)    | 0.079*<br>(0.047)            | 0.020<br>(0.045)    |
| Total #blogs posted              | −0.000<br>(0.000)   | 0.000<br>(0.000)    | −0.000<br>(0.000)            | −0.000<br>(0.000)   |
| Total #fans                      | 0.000<br>(0.000)    | 0.000<br>(0.000)    | 0.000<br>(0.000)             | 0.000<br>(0.000)    |
| Total #followings                | 0.000<br>(0.000)    | −0.000<br>(0.000)   | 0.000<br>(0.000)             | −0.000<br>(0.000)   |
| Total #reposts received          | 0.000<br>(0.002)    | −0.000<br>(0.002)   | −0.003<br>(0.004)            | −0.003<br>(0.004)   |
| Total #comments received         | 0.001<br>(0.000)    | 0.001<br>(0.000)    | 0.001<br>(0.001)             | 0.001<br>(0.001)    |
| Total #“likes” received          | −0.001<br>(0.001)   | −0.001<br>(0.001)   | −0.001<br>(0.001)            | −0.001<br>(0.001)   |
| Max. #reposts a blog received    | 0.019<br>(0.034)    | 0.016<br>(0.033)    | 0.050<br>(0.045)             | 0.058<br>(0.045)    |
| Max. #comments a blog received   | 0.000<br>(0.027)    | −0.015<br>(0.026)   | 0.007<br>(0.028)             | −0.013<br>(0.028)   |

(continued on next page)

**Table B.3** (continued).

| DV: Default Delinquency = 1   | Immediate emotions |                   | Average emotional tendencies |                   |
|-------------------------------|--------------------|-------------------|------------------------------|-------------------|
|                               | Anger              | Happiness         | Anger                        | Happiness         |
|                               | (1)                | (2)               | (3)                          | (4)               |
| Max. #“likes” a blog received | −0.009<br>(0.041)  | −0.005<br>(0.039) | −0.020<br>(0.044)            | −0.010<br>(0.041) |
| Constant                      | 0.115<br>(2.397)   | −0.794<br>(2.215) | −0.914<br>(2.430)            | −0.134<br>(2.442) |
| Loan approval month FE        | Yes                | Yes               | Yes                          | Yes               |
| Borrowers’ living province FE | Yes                | Yes               | Yes                          | Yes               |
| Mean of Emotion_m             | 0.163              | 0.269             | 0.166                        | 0.396             |
| Adj. R-squared                | 0.213              | 0.289             | 0.189                        | 0.202             |
| N                             | 110                | 110               | 110                          | 110               |

<sup>a</sup> PosExp and NegCons stand for positive-expectation group and negative-consequence group, respectively; neutral (control) group is used as the baseline group. For Income level, the category Income level = 1 serves as the baseline.

<sup>b</sup> Models (1) and (3) correspond to using anger as independent variables in Emotion\_m; Models (2) and (4) correspond to happiness. Suffix “m” stands for emotions at the pre-maturity stage.

<sup>c</sup> Standard errors are in parentheses. \*\*\*, \*\* and \* represent statistical significance at the 1%, 5% and 10% levels, respectively.

<sup>d</sup> Borrowers with an income level of 8 or 9 are not delinquent in our sample; Marital status is dropped due to collinearity.

**Table B.4**

A LLM approach for robustness check: Emotions at the pre-approval stage and loan default.

| DV: Default               | Immediate emotions  |                      | Average emotional tendencies |                     |
|---------------------------|---------------------|----------------------|------------------------------|---------------------|
|                           | Anger               | Happiness            | Anger                        | Happiness           |
|                           | (1)                 | (2)                  | (3)                          | (4)                 |
| Emotion_a                 | −0.273**<br>(0.137) | 0.130<br>(0.156)     | −0.324**<br>(0.161)          | 0.001<br>(0.220)    |
| Emotion_a_square          | 0.401**<br>(0.203)  | −0.222<br>(0.222)    | 0.534**<br>(0.269)           | −0.197<br>(0.372)   |
| Constant                  | −0.730**<br>(0.294) | −0.774***<br>(0.294) | −0.732**<br>(0.294)          | −0.746**<br>(0.295) |
| Controls                  | Yes                 | Yes                  | Yes                          | Yes                 |
| Loan application month FE | Yes                 | Yes                  | Yes                          | Yes                 |
| Living province FE        | Yes                 | Yes                  | Yes                          | Yes                 |
| Adj. R-squared            | 0.043               | 0.042                | 0.044                        | 0.042               |
| N                         | 1943                | 1943                 | 1943                         | 1943                |

<sup>a</sup> Models (1), (2), (5), and (6) correspond to using anger as independent variables in Emotion\_a and Emotion\_a\_square; Models (3), (4), (7), and (8) use happiness. Suffix “a” stands for emotions at the pre-approval stage.

<sup>b</sup> Standard errors are in parentheses. \*\*\*, \*\* and \* represent statistical significance at the 1%, 5% and 10% levels.

<sup>c</sup> Control variables include borrower characteristics, loan attributes, and microblog presence.

**Table B.5**

A LLM approach for robustness check: emotions at the pre-maturity stage and loan default.

| DV: Default Delinquency = 1   | Immediate emotions |                   | Average emotional tendencies |                     |
|-------------------------------|--------------------|-------------------|------------------------------|---------------------|
|                               | Anger              | Happiness         | Anger                        | Happiness           |
|                               | (1)                | (2)               | (3)                          | (4)                 |
| Emotion_m                     | −0.080<br>(0.466)  | 0.203<br>(0.573)  | −0.640<br>(0.730)            | −1.121**<br>(0.561) |
| Emotion_m_square              | 0.020<br>(0.733)   | −1.105<br>(0.859) | 1.364<br>(1.452)             | 0.843**<br>(0.425)  |
| Constant                      | −0.844<br>(0.847)  | −0.697<br>(0.858) | −0.692<br>(0.850)            | −0.529<br>(0.865)   |
| Controls                      | Yes                | Yes               | Yes                          | Yes                 |
| Loan application month FE     | Yes                | Yes               | Yes                          | Yes                 |
| Borrowers’ living province FE | Yes                | Yes               | Yes                          | Yes                 |
| Adj. R-squared                | 0.122              | 0.121             | 0.118                        | 0.115               |
| N                             | 437                | 437               | 437                          | 437                 |

<sup>a</sup> Models (1), (2), (5), and (6) correspond to using anger as independent variables in Emotion\_m and Emotion\_m\_square; Models (3), (4), (7), and (8) correspond to happiness. Suffix “m” stands for emotions at the pre-maturity stage.

<sup>b</sup> Standard errors are in parentheses. \*\*\*, \*\* and \* represent statistical significance at the 1%, 5% and 10% levels.

<sup>c</sup> Control variables include borrower characteristics, loan attributes, and microblog presence.

Table B.6

A LLM approach for robustness check: repayment reminders and emotions at the pre-maturity stage.

| DV: Default Delinquency = 1   | Immediate emotions  |                    | Average emotional tendencies |                     |
|-------------------------------|---------------------|--------------------|------------------------------|---------------------|
|                               | Anger               | Happiness          | Anger                        | Happiness           |
|                               | (1)                 | (2)                | (3)                          | (4)                 |
| PosExp · Emotion_m            | −1.522**<br>(0.773) | −0.679<br>(1.741)  | −1.216**<br>(0.615)          | −2.688**<br>(1.364) |
| NegCons · Emotion_m           | −1.485<br>(1.495)   | 3.936*<br>(2.063)  | −0.366<br>(1.549)            | 0.206<br>(1.589)    |
| PosExp                        | 0.237<br>(0.233)    | 0.603<br>(0.380)   | 0.261<br>(0.274)             | 0.930*<br>(0.548)   |
| NegCons                       | 0.256<br>(0.258)    | −0.761*<br>(0.418) | 0.100<br>(0.321)             | −0.383<br>(0.476)   |
| Emotion_m                     | 0.424<br>(1.079)    | −1.435<br>(1.165)  | 0.268<br>(0.937)             | 1.843<br>(1.183)    |
| Constant                      | −0.951<br>(2.334)   | −2.772<br>(2.300)  | −1.589<br>(2.450)            | −0.961<br>(2.384)   |
| Controls                      | Yes                 | Yes                | Yes                          | Yes                 |
| Loan approval month FE        | Yes                 | Yes                | Yes                          | Yes                 |
| Borrowers' living province FE | Yes                 | Yes                | Yes                          | Yes                 |
| Adj. R-squared                | 0.215               | 0.249              | 0.183                        | 0.217               |
| N                             | 110                 | 110                | 110                          | 110                 |

<sup>a</sup> PosExp and NegCons stand for positive-expectation group and negative-consequence group, respectively; neutral (control) group is used as a baseline.

<sup>b</sup> Models (1) and (3) correspond to using anger as independent variables in Emotion\_m; Models (2) and (4) correspond to happiness. Suffix “m” stands for emotions at the pre-maturity stage.

<sup>c</sup> Standard errors are in parentheses. \*\*\*, \*\* and \* represent statistical significance at the 1%, 5% and 10% levels, respectively.

<sup>d</sup> Control variables include borrower characteristics, loan attributes, and microblog presence.

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